

Research Paper

Long-term energy demand modeling and optimal evaluation of solutions to renewable energy deployment barriers in Benin: A LEAP-MCDM approach

Romain Akpahou^{a,*}, Lena D. Mensah^{a,c}, David A. Quansah^{a,c}, Francis Kemausuor^{b,c}

^a Department of Mechanical Engineering, Kwame Nkrumah University of Science and Technology (KNUST), Ghana

^b Department of Agricultural and Biosystems Engineering, Kwame Nkrumah University of Science and Technology (KNUST), Ghana

^c The Brew-Hammond Energy Centre, Kwame Nkrumah University of Science and Technology (KNUST), Ghana

ARTICLE INFO

Keywords:

Benin
LEAP
sustainable development
GHG emissions
energy demand

ABSTRACT

Approximately 60 % of Beninese lack access to electricity, relying heavily on traditional energy sources, electricity imports, and low renewable energy integration. This study aims to forecast the energy demand for Benin while reducing greenhouse gas (GHG) emissions and propose alternative solutions to clean energy deployment barriers. The Low Emissions Analysis Platform (LEAP) is used to explore the future energy demand for Benin and associated GHG emissions. Four scenarios have been developed, namely Business as Usual (BAU), High Economic Development (HED), Electrification and Urbanization Growth (EUG), and High Efficient Energy Application (HEEA). In addition, the Fuzzy AHP and TOPSIS techniques have been employed to rank and prioritize alternative solutions to Renewable Energy (RE) deployment barriers in the country. A total of eighteen sub-barriers were identified and classified into five main categories under the four pillars of sustainability (technical, social, economic, and environmental), and nine strategies were identified and proposed to overcome the barriers. The results show that under the BAU scenario, the total energy demand is expected to reach 445 PJ in 2050 from the current demand of around 164 PJ. A total GHG emission of 21 MtCO_{2e} is estimated under the BAU scenario in 2050. However, the HEEA scenario indicates that energy demand would considerably decrease resulting in low GHG emissions from the energy sector. The findings from the Fuzzy AHP ranking show that technical barriers, political and regulatory barriers, and financial and economic barriers, are the top three barriers hindering RE deployment in Benin. Moreover, adequate policy and regulatory framework emerges as the most weighted solution to eliminating barriers to RE deployment in the country. These results encourage the application of effective policy to support sustainable energy transition. The study's approach is applicable to other developing countries outside Benin, offering insights into RE barriers and potential alternative solutions.

List of abbreviations and acronyms: ABERME, Beninese Agency for Rural Electrification and Energy Control; BAU, Business As Usual; CO₂, Carbon dioxide; CSP, Concentrated Solar Power; DGRE, Directorate General Energy Resources; EUG, Electrification and Urbanization Growth; GDP, Gross Domestic Product; GDIZ, Gbongbong Industrial Zone; GHG, Greenhouse Gas; HEEA, High Efficient Energy Application; HED, High Economic Development; HH, Household; IEA, International Energy Agency; IRENA, International Renewable Energy Agency; INSAE, National Institute of Statistics and Economic Analysis; INStAD, National Institute of Statistics and Demography; LEAP, Low Emission Analysis Platform; LPG, Liquefied Petroleum Gas; MCVDD, Ministry of Living Environment and Sustainable Development; MESSAGE, Model for Energy Supply Strategy Alternatives and their General Environmental Impacts; MtCO_{2e}, Million metric tonnes of CO₂ equivalent; NEMO, Next Energy Modeling System for Optimization; NDC, Nationally Determined Contribution; OSeMOSYS, Open Source Energy Modeling SYSTEM; PJ, Petajoules; PRODERE, Renewable Energy Development Program; PROVES, Solar Energy Promotion Project; PONADER, National Renewable Development Policy; PANEE, National Energy Efficiency Action Plan; PV, Photovoltaic; SBEE, Beninese Company of Electrical Energy; SIMPACTS, Simplified approach to estimating electricity generation's external costs and impacts; SDG7, Sustainable Development Goal 7; SSA, Sub-Saharan Africa; SIE-Benin, Energy Information System of Benin; UNDP, United Nations Development Programme; AHP, Analytic Hierarchy Process; TOPSIS, Technique for Order of Preference by Similarity to Ideal Solution; MCDM, Multi-criteria Decision Making; CC_N, Closeness Coefficient.

* Corresponding author.

E-mail address: romain.akpahou@aims.ac.rw (R. Akpahou).

<https://doi.org/10.1016/j.egy.2024.07.055>

Received 20 April 2024; Received in revised form 16 July 2024; Accepted 27 July 2024

Available online 10 August 2024

2352-4847/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Electricity is a widely used secondary form of energy in many sectors. However, over half a billion people in the world still lack access to electricity, with Sub-Saharan Africa (SSA) having the largest access deficit (IEA et al., 2021). Given this, ensuring universal access to clean and affordable energy services to communities, as stated in Sustainable Development Goal 7 (SDG7), has become a top priority for policy-makers, businesses, and governments worldwide (United Nations, 2018).

Benin, a developing country in SSA, has an energy transition program that aims to provide its populace access to clean and affordable energy services (Sotiman Yotto et al., 2021). In 2020, Benin had a Gross Domestic Product (GDP) of \$USD 15.65 billion with a growth rate of 2.7 %/year compared to 2019 (World population review, 2023). According to Mensah et al. (2022), the country's economic growth and electric consumption correlate. However, Benin depends on its neighboring countries for electricity supply and has low access to electricity, particularly in rural communities (Sotiman Yotto et al., 2021).

Despite the abundant energy resources available in Benin (Adamon et al., 2020), the power sector is still subject to regular power cuts and recurrent electricity crises (Akpahou, Odoi-yorke, et al., 2023). Most of the country's electricity supply is imported from Ghana, Nigeria, and Togo (Mensah et al., 2022). However, the country's considerable untapped renewable energy resources can be harnessed to fill the gap between demand and supply and provide access to electricity for the populace (Odou et al., 2020). In 2018, Benin's total energy consumption was 17.4 GJ/capita (SIE-Benin, 2021), which was notably lower than that of its neighboring country Ghana, which consumed 10.8 TJ/capita of energy during the same period (Energy Commission Ghana, 2020).

The predominant share of energy consumption in Benin is directed towards the transport and residential sectors, emphasizing their central role in overall energy utilization. Conversely, the distribution of the remaining energy is spread among services, industry, and agriculture. Notably, despite the pivotal roles played by the industry and agriculture sectors in creating added value, generating employment, and fostering economic growth, their combined contribution to final energy consumption is modest, accounting for approximately 4.2 %. This relatively stagnant sectoral distribution from 2010 to 2020 underscores the persistent challenges faced by industries and agricultural activities in significantly impacting the overall energy landscape (SIE-Benin, 2021).

Particularly intriguing is the substantial energy demand within households, largely attributed to the widespread use of firewood and charcoal for cooking purposes. This prevalent energy consumption pattern mirrors the energy profiles commonly observed in non-industrialized countries where a significant portion of the energy is directed towards domestic activities and transportation, as evidenced in the detailed breakdown provided in Table 1. This insight not only sheds light on the dynamics of energy utilization in the context of household practices but also accentuates the distinct characteristics of energy consumption in regions characterized by a non-industrialized socioeconomic framework (MCVDD, 2021). Further exploration of these trends could offer valuable insights into devising targeted strategies for optimizing energy consumption and fostering sustainable practices in these sectors.

Table 1
Final energy consumption in Benin from 2017 to 2020 (SIE-Benin, 2021).

Sector \Year	Final energy consumption (PJ)			
	2017	2018	2019	2020
Industry	2.49	5.32	6.12	6.44
Transport	68.48	69.32	57.74	58.91
Households	69.1	70.76	73.05	75.41
Services	15.09	15.48	15.92	16.02
Agriculture	0.46	0.48	0.46	0.47

Fig. 1 shows the mix electric of Benin from 2016 to 2020. In 2020, natural gas emerged as the primary energy source, constituting a substantial 71.63 % of the total mix. Meanwhile, solar photovoltaic (PV) contributed a modest 0.30 %, contrasting with its 1.36 % share in 2016. The surge in energy demand, driven by population growth and diverse activities, has marked recent decades in Benin. Benin's electricity supply relies on two primary sources: national production and imports. The Electricity Community of Benin (CEB), a collaborative effort between Benin and Togo, plays a crucial role in delivering electrical power to Benin (Odou et al., 2020). The national component within CEB is managed by the government-owned national power supply corporation, SBEE. To meet the electricity demand, SBEE procures a significant portion of its energy from CEB (Sotiman Yotto et al., 2021). Importantly, CEB sources a substantial part of its energy from imports, primarily from Ghana and Nigeria. Additionally, it leverages its production facilities, including the Nangbéto hydroelectric power station situated on the Mono River, to contribute to Benin's energy needs.

As of 2020, Benin's national electricity generation capacity was only 174.5 MW. The largest thermal power plant is located at Maria Gléta, with a capacity of 127 MW (Sotiman Yotto et al., 2021). Traditional biomass and petroleum products meet most of Benin's energy demand (SIE-Benin, 2021). However, recognizing the importance of increasing access to electricity while diversifying the total energy mix, the government of Benin has planned to increase the share of renewable in its total energy mix (Akpahou and Odoi-yorke, 2023). To achieve these goals, the government has implemented several initiatives, such as the Rural Electrification and Renewable Energy Project, the Solar Energy Promotion Project (PROVES), and the Renewable Energy Development Program (PRODERE) (Odou et al., 2020). Additionally, Benin has significant potential for solar, biomass, wind, and hydropower, which could be harnessed to boost the country's energy supply and reduce reliance on traditional fuel-based resources (Mensah et al., 2022). Despite this, the share of RE resources in the final supply mix is just at 3.5 %. Therefore, adopting appropriate energy planning strategies is crucial to prevent an imbalance between future energy demand and supply in the country. It has become primordial to identify the main barriers that hinder the deployment of RE in the Benin Republic and propose appropriate solutions to overcome the identified barriers. In this context, this study aims to first forecast the energy demand and associated GHG emissions and next, explore and dissect the multifaceted barriers that hamper the development of RE, shedding light on the complexities and nuances that decision-makers, researchers, and stakeholders should consider to accelerate the global transition to sustainable energy sources.

This study first utilizes the LEAP model to forecast the Benin Republic's future energy demand and secondly, proposes an integrated MCDM approach to evaluate and rank alternative solutions to eliminate RE barriers to accelerate clean energy development in the country. The present study is novel in its approach and perspective of identification, assessment, and analysis of RE barriers in developing economies, particularly in Benin. It contributes in many ways: (1) forecast the future

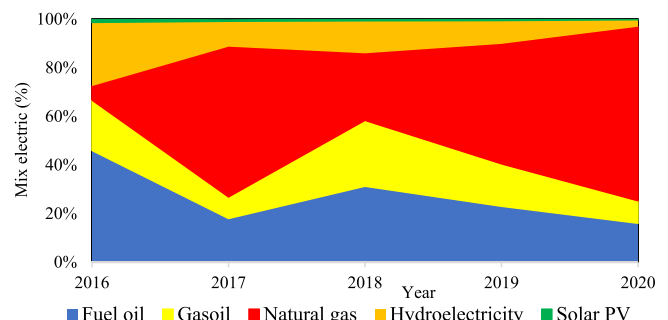


Fig. 1. Share of electricity mix of Benin from 2016 to 2020 (SIE-Benin, 2021).

energy demand of the Benin Republic till 2050 using the LEAP model. (2) analyze different scenarios and recommend strategies that will satisfy Benin's long-term energy requirements while reducing GHG emissions. (3) identify, assess, evaluate, and prioritize barriers to RE deployment in Benin by using the Fussy AHP method, (4) examine strategies to eliminate the RE barriers using the Fuzzy TOPSIS technique. The novelty of the study is also extended towards contributing to the knowledge of barriers to RE deployment and how to tackle them to achieve the country's objectives under the Paris Agreement. The approach in this study can represent and highly relate to other less developed and developing countries outside Benin where the outputs and policy implications can indirectly be used to explore national and regional RE barriers and alternative solutions. The remaining sections of this study are outlined as follows: [Section 2](#) presents a literature review on energy demand forecasting and MCDM applications for evaluating barriers to RE deployment, and [Section 3](#) presents the methods and materials utilized, including a comprehensive overview of the LEAP simulation tool and MCDM methods. Subsequently, [Section 3](#) highlights the results and discussions, while [Section 4](#) provides conclusions and recommendations.

2. Literature review

This section presents a comprehensive review of the literature on energy demand modeling and Multi-Criteria Decision Making approach for prioritizing barriers to RE technology deployment.

2.1. Energy demand forecasting

Globally, several authors have extensively applied energy planning models to forecast energy demand and decarbonization pathways in different countries. For instance, [Neniškis et al. \(2021\)](#) used the Model for Energy Supply Strategy Alternatives and their General Environmental Impact (MESSAGE) to improve the Lithuanian transport sector. The proposed approach provided 7 % lower emissions for the year 2018 than statistical data estimation. Additionally, [Adeyemi-Kayode et al. \(2022\)](#) employed the long-range energy alternative plan (LEAP) model to evaluate greenhouse gas (GHG) mitigation plans and future sustainable energy generation mix in Nigeria. It was found that integrating renewable energy, natural gas, and biomass in the energy production plan is the best scenario for the energy transition in Nigeria. Likewise, [Mekonnen et al. \(2022\)](#) used the open-source energy modeling system (OSeMOSYS) to forecast the energy transition in Ethiopia. The authors found that the country should prioritize transitioning from fossil fuel to renewable energy resources such as wind and solar to develop a decarbonized future electricity generation system.

LEAP is a widely used simulation tool for energy demand forecasting and decarbonization pathways analysis for most developing countries ([Huang et al., 2011](#); [Machado et al., 2019](#); [Mahboob et al., 2021](#); [Syed et al., 2014](#)). For example, [Wambui et al. \(2022\)](#) applied LEAP, and Next Energy Modeling System for Optimization (NEMO) models to investigate the impact of alternative electricity generation pathways on GHG emissions and system cost reduction in Kenya. The findings showed that integrating more renewable energy sources has the potential to significantly reduce overall system costs and GHG emissions. [Perwez et al., \(2015\)](#) employed LEAP to develop long-term electricity demand and supply scenarios in Pakistan. The results showed cost reductions and GHG emissions reductions in the country in the presence of adequate policy development.

[Gebremeskel et al. \(2023\)](#) used the OSeMOSYS-LEAP soft-linking modeling approach for Ethiopia's long-term electricity supply and demand modeling. The authors revealed that Ethiopia needs to increase its share of renewable energy sources such as solar photovoltaic (PV), concentrated solar power (CSP), wind, and geothermal to fill the gap between demand and supply in the country. [Mondal et al. \(2010\)](#) applied LEAP to analyze future energy demand scenarios for

Bangladesh. It was observed that the model is suitable for energy demand forecasting in the country. Also, the findings have important implications for energy conservation and the country's economic development.

Furthermore, [Emodi et al. \(2017\)](#) used LEAP to analyze the impacts of energy policies on low-carbon development in Nigeria. The authors recommended that Nigeria explores policies relevant to energy demand and CO₂ emissions reduction by prioritizing renewable energy sources. [Raza et al. \(2022\)](#) employed LEAP to develop an energy demand forecasting model between 2018 and 2030 for Pakistan. The study focused on carbon emissions (CO₂) reduction analysis with recommendations on integrating local renewable energy sources to mitigate Pakistan's energy crisis. In addition, [Gebremeskel et al. \(2021\)](#) utilized LEAP to perform long-term electricity demand forecasting in Ethiopia under different scenarios. The authors suggested that the Ethiopian government should develop long-term energy policies and strategies to positively impact the country's energy sector, focusing on CO₂ mitigation measures. Additionally, LEAP has been employed in Pakistan to investigate 100 % RE integration to analyze cost reduction and associated emissions ([Hussain et al., 2021](#)). The studies reviewed above highlight the use of energy modeling tools, particularly in developing countries context for energy demand forecasting.

2.2. Evaluation of barriers and solution to RE deployment with MCDM

Currently, more than half of Benin's population lacks access to electricity and the electricity produced from RE sources represents only 3.5 % of the total electricity supply. Therefore, it is important to identify and prioritize the barriers to RE deployment in Benin and propose alternative solutions and strategies that would accelerate the transition to clean and sustainable energy technologies in Benin Republic.

Multicriteria decision-making (MCDM) approaches are important in determining the weights and ranking different criteria to make objective decisions. Several MCDMs exist in the literature and have been applied in many countries to identify and rank barriers to RE technologies ([Ghimire and Kim, 2018](#)). For instance, [Solangi et al., \(2021\)](#) conducted an assessment of barriers to RE development in Pakistan to identify challenges limiting the transition to RE in the country. Using the Analytical Hierarchy Process (AHP) combined with the fuzzy TOPSIS method, the authors identified seven barriers and twenty-two sub-barriers and proposed strategies to overcome some of the barriers. The results show that market, Economic and financial, and policy are the main barriers to RE development in the region. [Gordon et al., \(2023\)](#) evaluated the social and technical barriers to domestic hydrogen development in the United Kingdom (UK) as a clean energy source. The findings of the study showed that future strategies should consider the interactions between technical, social, market, and political dimensions of green hydrogen transition.

In addition, [Lee et al., \(2022\)](#) identified barriers and strategies to green hydrogen production for RE energy technology development in South Korea. The authors employed the expert Delphi surveys approach and AHP technique to identify five factors and fifteen sub-factors that eradicate the development of clean hydrogen in the country. Institutional and political factors in addition to the high initial installation cost associated with green hydrogen generation are the most important barriers identified. By employing a combined MCDM approach, [Rahimirad and Sadabadi, \(2023\)](#) assessed green hydrogen technology development in the Iranian context. The authors identified several barriers and sub-barriers and proposed alternatives to eliminate the barriers.

[Çelikbilek and Tüysüz, \(2016\)](#) employed an integrated MCDM approach to evaluate RE sources based on experts' points of view. Grey DEMATEL, grey ANP, and grey VIKOR are the applied MCDM approaches considering Turkey as a region of interest. The findings demonstrate the applicability of MCDM approaches in evaluating RE sources to accelerate the transition towards sustainable energy sources.

In India, a Fuzzy AHP approach was used to prioritize barriers to offshore wind energy development [Dhingra et al., \(2022\)](#). The authors developed a framework to identify and categorize barriers and the results indicate that the lack of financial resources, inadequate policy, and grid-connection challenges are some of the main barriers to the development of offshore wind technologies in the country. Similarly, [Ghimire and Kim, \(2018\)](#) analyzed barriers to RE development from the Nepalese perspective using the AHP approach. Twenty-two barriers were identified and classified under social, political, technical, economic, administrative, and geographic types of barriers. The findings show that the economic and political barriers are the most important challenges that limit the development of RE in Nepal.

Furthermore, [Maman Ali and Yu, \(2021\)](#) assessed the impact of RE policy implementation on sustainable energy development in the Economic Community of West African States (ECOWAS). The results demonstrate that further policy and reform are needed in the region to support the development of strategies to achieve a long-term sustainable energy transition. When answering the question on the barriers to RE entrepreneurs in developing countries, [Gabriel, \(2016\)](#) identified six key barriers including the lack of skilled labor, lack of financial resources, the price of RE technology, lack of adequate infrastructure, inadequate governmental policy, and dominance of incumbents. [Sen and Ganguly, \(2017\)](#) analyzed the opportunities, barriers, and issues associated with the development of RE technologies. The authors have identified several barriers which include market failures, informational and awareness barriers, and policy and socio-cultural barriers. Several alternatives and strategies have been recommended for a clean and sustainable energy transition.

[Wyllie et al., \(2018\)](#) have identified the main barriers to deploying alternative energy sources in Barbados. The challenges associated with the development of solar energy resources have been investigated and the results showed that inadequate policy and lack of incentives are significant barriers to RE development in the region. In addition, an assessment of barriers and opportunities to RE development has been conducted in Chile [Nasirov et al., \(2016\)](#). The authors employed the AHP technique to rank barriers based on experts' opinions. The results indicate that regulatory and institutional barriers and technical and infrastructure matters are the top two sets of barriers to RE development in the region. [Numata et al., \(2020\)](#) analyzed the barriers to developing mini-grid RE resources in Myanmar using the AHP technique.

[Shahzad et al., \(2022\)](#) have identified and classified RE barriers in developing RE entrepreneurship in Pakistan. The authors employed the Pythagorean fuzzy AHP method to estimate weights and rank twelve barriers. The findings indicate that the legal and regulatory framework is the first and principal obstacle to RE deployment in the country. AHP and fuzzy TOPSIS approach are used in Malawi to evaluate barriers and propose strategies to accelerate RE deployment [Chisale and Lee, \(2023\)](#). Six main barriers and nineteen sub-barriers have been identified and the result indicates that economic and lack of governmental policy are the top barriers. [Asante et al., \(2022\)](#) also combined the CRITIC method with the fuzzy TOPSIS approach to prioritize strategies that could help eliminate barriers to RE development in Ghana. Lack of RE policy, technical challenges, lack of funding mechanism, and adequate information are some of the main barriers to RE development in Benin Republic [Sotiman Yotto et al., \(2021\)](#).

The literature review has demonstrated the application of MCDM to evaluating barriers and proposing solutions to accelerate clean energy transition. However, to the best of the authors' knowledge, none of the studies have investigated scenarios for long-term energy demand modeling in Benin. In addition, there is a gap in efficiently prioritizing adequate solutions to RE deployment barriers and the action needed to boost clean energy transition in a developing region like Benin. Therefore, this study employed an integrated MCDM approach, combining fuzzy AHP and fuzzy TOPSIS methods to calculate the weights and rank the identified barriers and solutions. This study is the first of its kind to be conducted in the Benin Republic and focuses specifically on the

country's energy demand and GHG emissions reduction scenarios for sustainable development plans. The modeling approach would offer valuable insights into sustainable energy policy development and has the potential for successful replication in other countries within the Sub-Saharan African region. The study's findings could help inform long-term energy demand planning and decarbonization pathways analysis in the Benin Republic and other developing countries by analyzing various scenarios.

3. Methods and material

This section presents the material and the methodological approach used in forecasting the future energy demand and evaluating barriers and alternative solutions to RE deployment in Benin.

3.1. Energy demand forecasting with LEAP

The Low Emissions Analysis Platform (LEAP), originally known as the Long-range Energy Alternatives Planning system, is a scenario-based energy and environment modeling tool that uses a comprehensible accounting approach to analyze how energy is used, transformed, and produced in a given economy based on various possible assumptions ([SEI, 2005](#)). It is a widely used energy modeling tool for forecasting energy demand, analyzing decarbonization pathways, and formulating energy policy ([Gebremeskel et al., 2021](#)). LEAP can solve local, national, and international energy planning issues ([Raza et al., 2022](#)).

This study uses the model to calculate the energy demand for different sectors in Benin up to 2050, considering 2017 as the base year for the model setup. The study selected 2017 as the base year because it is the most recent year with the maximum data availability. Data on households that had access to electricity and the percentage share of combustible used for cooking and lighting in rural and urban areas was obtained from a demographic and health survey conducted by the National Institute of Statistics and Economic Analysis (INSAE) in Benin in 2017 ([INSAE and I.C.F., 2019](#)). The LEAP modeling approach is based on a bottom-up approach that combines energy demand with urbanization, GDP, and technological changes and helps policymakers develop appropriate energy planning policies ([Mondal et al., 2010](#)). In this study, the energy demand model focused on five major sectors in Benin: agriculture, transport, industry, households (HH), and services.

The LEAP model uses [Eq. \(1\)](#) to compute energy demand ([Wambui et al., 2022](#)):

$$D_{b,s,t} = TA_{b,s,t} \times EI_{b,s,t} \quad (1)$$

Where D is the energy demand, TA is the total activity, EI is the energy intensity, and b, s, and t denote branch, scenario, and year, respectively. In addition, the carbon emission from final energy consumption is estimated using [Eq. \(2\)](#) ([Ouedraogo, 2017](#)):

$$CE_{b,s,t} = TA_{b,s,t} \times EI_{b,s,t} \times EF_{b,s,t} \quad (2)$$

Where CE is the carbon emissions, and EF is the carbon emissions factor from a sector or branch. This study adopted the default emission factors in the LEAP model suggested by the Intergovernmental Panel on Climate Change (IPCC) suitable for climate change mitigation analyses ([SEI, 2005](#)). In addition, a grid electricity-specific emission factor of 0.439 tCO₂/MWh ([Sam-Amobi et al., 2019](#)) for Nigeria is used for electricity import emissions estimation, since Benin imports the majority of its electricity from Nigeria. Emissions from wood are not treated as carbon-neutral even though the government is making an effort to tackle deforestation by planting trees.

The LEAP-Benin modeling approach is based on different levels: sector level (households, agriculture, industry, transport, and service), sub-sector level (rural, urban), sub-sub sector (% electrified, % non-electrified in each sub-sector), and end-use which includes lighting, cooking, refrigeration, and others use. Energy intensity values based on

fuel types and energy consumption for different devices are included in the energy demand analysis to estimate energy requirements at each level. However, due to limited data availability, in the industry, commercial, transport, and agriculture sectors, energy use was directly divided into a share of energy use, including residual fuel oil, LPG, coal, biomass, electricity, etc. Data on the share of energy use for different activity sectors in Benin between 2016 and 2020 was obtained from [SIE-Benin, \(2021\)](#). This study used data for the year 2017 base year. The structure of the LEAP-Benin modeling approach for the household sector is shown in [Fig. 2](#).

3.1.1. LEAP model data and assumptions

This study selected relevant data sources for the data collection on electricity consumption, GDP, electrification rate, number of households, population growth, etc. Data were mainly collected from the electric power sector of Benin, including the Ministry of Energy (ME), the Beninese Company of Electrical Energy (SBEE), the General Directorate of Energy Resources (DGRE), and the Beninese Agency for Rural Electrification and Energy Control (ABERME) ([MCVDD, 2021](#); [Ministry of Energy, 2020](#); [SIE-Benin, 2021](#)). Relevant published papers in developing countries, including international reports, were also assessed ([Emodi et al., 2017](#); [Global Data, 2023](#); [Raza et al., 2022](#); [SE4ALL, 2023](#); [World bank, 2023](#)). A report by [INSTaD, \(2022\)](#) has already projected Benin's population from 2014 to 2063.

According to the [World Bank \(2023\)](#), Benin's population in 2017 was about 11,596,776 with an average household size of 6 inhabitants ([Yatokpa, 2015](#)). Benin's primary energy supply comprises biomass energy, mineral coal, petroleum products, natural gas, and electricity. In 2017, the primary energy supply was estimated at 194.5 PJ, with the final energy consumption at 155.9 PJ. Losses related to processing and distribution equal to 38.6 PJ. Households and transport alone accounted for more than 88 % of the final energy consumption. Household consumption is dominated by biomass, while the transport sector

exclusively uses petroleum products ([SIE-Benin, 2021](#)).

Households sector

In the households sector, biomass constitutes nearly 95.5 % of the share, while petroleum products and electricity are in very low quantities, accounting for 2.28 % and 1.66 %, respectively ([Ministry of Energy, 2020](#)). The excessive use of wood can be attributed to significantly lower costs compared to commercial energy sources. In this sector, electricity usage addresses essential needs such as lighting, food refrigeration, air conditioning (with fans or air conditioners depending on household class), and entertainment, including television, radio, mobile phones, and other electronic devices.

According to the [Ministry of Energy \(2020\)](#), about 55.4 % of Beninese lived in rural areas in 2017. Furthermore, electricity access in the country was very low, at 34.5 % ([World bank, 2023](#)), with 54.2 % access in urban areas and 17.2 % in rural areas. However, Benin's households' electricity access is slightly different from the population's electricity access rate. A report by [INSAE and ICF \(2019\)](#), revealed that about 55.5 % of urban households (HH) in the country have access to electricity, while only 20.5 % of HH in rural areas are connected to the electricity grid. Different type of fuel is used in the country for cooking.

[Table 2](#) presents the percentage share of combustibles used in Benin ([INSAE and ICF, 2019](#)). Furthermore, in Benin, all electrified urban and rural households used electricity for lighting. The non-electrified rural area is characterized by the use of flashlights and kerosene lamps for lighting, accounting for around 72.9 % of the energy used for lighting ([Ministry of Energy, 2020](#); [SIE-Benin, 2021](#)).

Transport sector

The transportation sector relies 100 % on fossil fuels. In Benin, the road sector consumes almost all of this energy, representing 45.9 % of the 50.2 % share that petroleum products account for in the final energy consumption in 2020. The consumption in the road sector is divided between diesel (38 %) and gasoline (62 %) ([Ministry of Energy, 2020](#); [PONAME, 2020](#)). The average consumption growth rate of the

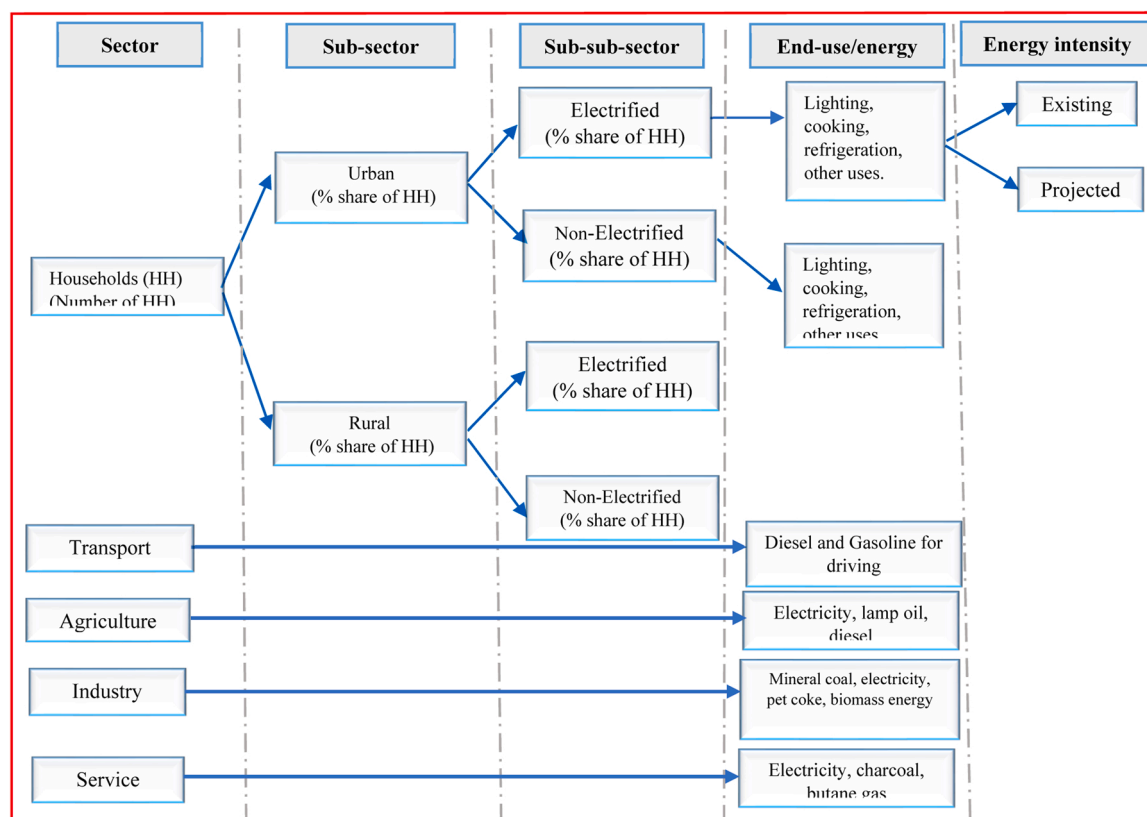


Fig. 2. LEAP-Benin structure for the different sectors.

Table 2

Electricity access and combustible used for cooking per household (%) (INSAE & ICF, 2019).

Household Characteristics	Households (HH)		
	Urban	Rural	Average
Electricity			
Yes	55.5	20.5	35.6
No	44.5	79.5	64.4
Combustible for cooking			
Charcoal	43.9	13.7	26.7
Firewood	39.9	78.5	61.9
LPG	9.7	0.9	4.7
Electricity	0.5	0.2	0.3
Lignite/coal	1.8	0.4	1
Kerosene	0.2	0.1	0.2
Agriculture residue (herbs, straws, branches)	4	6.2	5.2

transportation sector over the last five years is 2 %.

Agriculture sector

Having changed status, consumption in the agricultural sector has experienced both structural and quantitative changes. The primary forms of energy consumed are diesel (88 %), electricity (8 %), and lamp oil (4 %) (MCVDD, 2021). Agriculture, along with industry, remains marginal in electrical consumption. Furthermore, the agricultural sector is not mechanized, indicating limited development in these sectors.

Industry sector

This sector is characterized by a significant share of mineral coal (41.22 %), electricity (14.59 %), and petroleum coke (14.49 %). The share of biomass energy is also quite substantial, representing nearly 8.25 % of the sector's consumption.

Service sector

In 2020, this sector consumed 16.02 PJ, with 58 % from firewood, 14 % from charcoal, 13 % from electricity, 2 % from other biomasses, and 1 % from butane gas. Small units (shops, supermarkets, bakeries, small agro-food processing units, etc.) account for 76.7 %, while hotels and restaurants represent 2.5 % of energy consumption. State-owned companies and offices account for only 2 % of the final energy consumption. However, electricity consumption in the sector increased at an annual growth rate of 7.4 %, but its relative share remained nearly stable between 2019 and 2020, comprising approximately 10.3 % of the total final consumption (Ministry of Energy, 2020).

Key assumptions adopted in the LEAP model are based on socio-economic and urbanization parameters, including GDP, population growth, urban and rural population, number of households, and other relevant data that affect the country's final energy consumption. Assumptions made in the case of this study based on the formulated scenarios are summarized in Table 3.

3.1.2. Scenarios development

In this study, a Business-as-Usual scenario (BAU) is developed. In addition, three different scenarios, namely High Economic Development (HED), Electrification and Urbanization growth (EUG), and High Energy Efficient Application (HEEA) were also developed.

Table 3

Key assumptions for LEAP-Benin under all scenarios (PONAME, 2020; SIE-Benin, 2021; Ouedraogo, 2017; Emodi et al., 2017; Gebremeskel et al., 2021).

Key assumptions	BAU			HED			EUG			HEEA		
	2017	2025	2050	2017	2030	2050	2017	2030	2050	2017	2030	2050
Households' growth rate (%/year)	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Urban electrification rate (%)	55.5	95	95	55.5	95	95	55.5	100	100	55.5	100	100
Rural electrification rate (%)	20.5	65	65	20.5	65	65	20.5	90	100	20.5	95.0	100
Urbanization rate (%)	44.6		60.0	44.6		60	44.6	70	80	44.6	75	85
Industry growth rate (%/year)	6.0		6.5	6.0		10	6.0		6.5	6.0		6.5
Agriculture growth rate (%/year)	6.0		6.5	6.0	6.5	9.0	6.0		6.5	6.0		6.5
Services growth rate (%/year)	5.1		5.6	5.1		7	5.1		5.6	5.1		5.6
Electric and clean cooking stoves penetration level (%)	5			5			5	20	40	5	25	50

The BAU scenario, also known as the reference scenario, represents the case with no governmental policy intervention. It is assumed that natural gas and biomass will still dominate Benin's primary energy consumption and that the current problems the energy sector faces will continue. According to the International Renewable Energy Agency (IRENA), only 41 % of Benin's population had access to electricity in 2022 (IRENA, 2022). The share of non-biomass renewable energy in Benin's total mix of electricity is estimated at 3.17 % in 2020 (including only solar PV and hydroelectricity) (SIE-Benin, 2021).

Therefore, in terms of electricity access in the country, the objective is to achieve urban and rural electrification rates of 95 % and 65 %, respectively, by 2025 (SE4ALL, 2023), and regarding renewable energy targets, the goal is to achieve a 24.6 % non-biomass renewable energy fraction in the total energy mix by 2025 (World bank, 2023). This target is considered in the BAU scenario, and future assumptions are made based on trends and judgments regarding GDP growth rates, agriculture growth rates, industry growth rates, etc. The data in the base year are collected from (INSAE and ICF, 2019; PONAME, 2020; SIE-Benin, 2021; IRENA, 2022), and similar work in West African developing countries (Emodi et al., 2017; Gebremeskel et al., 2021; Mensah et al., 2023; Ouedraogo, 2017). Energy demand projections are made considering industry and commercial parks in operation, under consideration, or planned to be developed in Benin. The Glo-Djigbé Industrial Zone in Benin is a remarkable project aimed at revolutionizing the industry and trade sector in the country in the coming years. It holds the key to transforming agricultural products like mahogany, cotton, and pineapple, thereby significantly boosting the nation's economy. Spanning approximately 1640 ha, this expansive zone caters to various needs including industrial, commercial, and storage, as well as common infrastructure and facilities (GDIZ, 2023). The assumptions underlying these projects consider their significant impact on the overall economic development of Benin.

The HED scenario builds up on the reference scenario and assumptions are made considering significant economic development in the country due to the implementation of many developmental plans. Industry activities are assumed to grow by 10 %/year in 2050, while commercial and agricultural activities are expected to grow by 7 %/year and 9 %/year, respectively in 2050. This is based on the government of Benin's plans and previous trends in Benin (PONAME, 2020; MCVDD, 2021), and patterns in West Africa (Emodi et al., 2017; Ouedraogo, 2017). Other parameters are kept as same as in the BAU scenario. This section is based on the BAU scenario and considers ambitious economic development in Benin by 2050.

The Electrification and Urbanization growth (EUG) scenario is formulated by assuming that the government has targets for reducing CO₂ emissions by developing more GHG reduction plans. It also considers integrating more renewable energy sources such as solar, small hydro, and bioenergy. Built under the BAU scenario, EUG assumed that the country would shift from biomass-based cooking stoves to more environmentally friendly cooking technology with more penetration of improved and clean cooking stoves.

The government of Benin aims to reduce cumulative GHG emissions in the energy sector by 11.51 % between 2021 and 2030 compared to

the current status quo as in its nationally determined contribution (NDC) plan (MCVDD, 2021). To achieve this, the government aims to use a combination of off-grid technologies and grid expansion. Improved cooking stoves are to be promoted for charcoal, wood, and other biomass. Integrating electric cars to replace fossil-based combustion engines by 2050 with a high urbanization rate per year is considered based on government ambitions.

An electrification rate of 100 % in urban areas by 2030 is assumed here as targeted in SE4ALL, (2023). In rural areas, the electrification rate is supposed to reach 90 % in 2030 and 100 % by 2050. An urbanization rate of 70 % and 80 % are assumed, respectively, for 2030 and 2050 based on previous trends and the government's targets (MCVDD, 2021; PONAME, 2020).

The last considered scenario is the **High Energy Efficient Application (HEEA)** scenario. This scenario is based on the EUG scenario but with major differences in country policy and GHG emissions reduction targets. Benin's specific goals for energy efficiency include eliminating inefficient incandescent lamps, reducing electricity network transmission and distribution losses, and achieving universal access to clean, affordable, efficient, and sustainable cooking for the entire population by 2030 (Yatokpa, 2015). Light-emitting diodes (LEDs) will be promoted, and by 2030, the rural electrification rate will reach 95 %. The urbanization rate will be higher than in the EUG scenario, reaching 75 % and 85 %, respectively, in 2030 and 2050.

Furthermore, the country is supposed to integrate improved cooking stoves in urban and rural areas based on the objectives submitted under the Paris Agreement (MCVDD, 2021). In the report, the objectives include promoting energy efficiency and utilization by providing improved stoves to 140,000 newly established households. Increasing access to household gas cooking equipment for an additional 275,000 families by subsidizing the purchase of small 6 kg gas equipment and burners by up to 30 %. The government intends also to electrify a total of 600 localities from 2021 to 2030 by connecting them to the power grid, thereby expanding access to electricity. Promoting access to electricity for 424,000 households that currently rely on kerosene for lighting in areas that will be connected to the Société Beninoise d'Énergie Electrique (SBEE) networks. These proposed initiatives aim to mitigate

environmental impacts, enhance energy efficiency, improve access to clean energy sources, and elevate the overall quality of life for Benin's population.

Energy demand projection from industry, households, agriculture, and transport sector is based on previous years' consumption patterns in Benin (SIE-Benin, 2021; PONAME, 2020) and trends in the West African region (Ouedraogo, 2017; Emodi et al., 2017).

3.2. MCDM approach to prioritizing alternative solutions to RE deployment barriers

3.2.1. Data collection and survey

Fig. 3 illustrates the general framework of the study. The authors first identified the main barriers to RE development in Benin through an in-depth review of the literature and expert consultation in the energy sector. A set of keywords was entered in databases such as Google Scholar, Science Direct, and Scopus for the barriers and strategies identification. In addition, a questionnaire is developed and presented to experts to identify and confirm barriers and strategies to accelerating clean energy integration in the country. A total of eighteen sub-barriers were identified and classified into five (05) main categories of barriers including Financial and Economic barriers, Socio-cultural barriers, Technical barriers, Political and Regulatory barriers, and geographical and environmental barriers. Moreover, nine (09) strategies were identified and proposed to overcome the barriers. Tables 4 and 5 present the identified barriers and the proposed alternatives to accelerate RE development in Benin.

To achieve the aim of this study, an integrated MCDM approach is used combining the Fuzzy-AHP approach, and fuzzy-TOPSIS technique. The fuzzy-AHP method employed is based on the Buckley approach and is used to evaluate weights and rank the barriers and sub-barriers. In addition, the fuzzy TOPSIS is employed to evaluate and prioritize the proposed strategies for decision-making.

3.2.2. Fuzzy AHP method

Fuzzy Analytic Hierarchy Process (F-AHP) is an advanced adaptation of the fundamental Analytic Hierarchy Process (AHP) originally

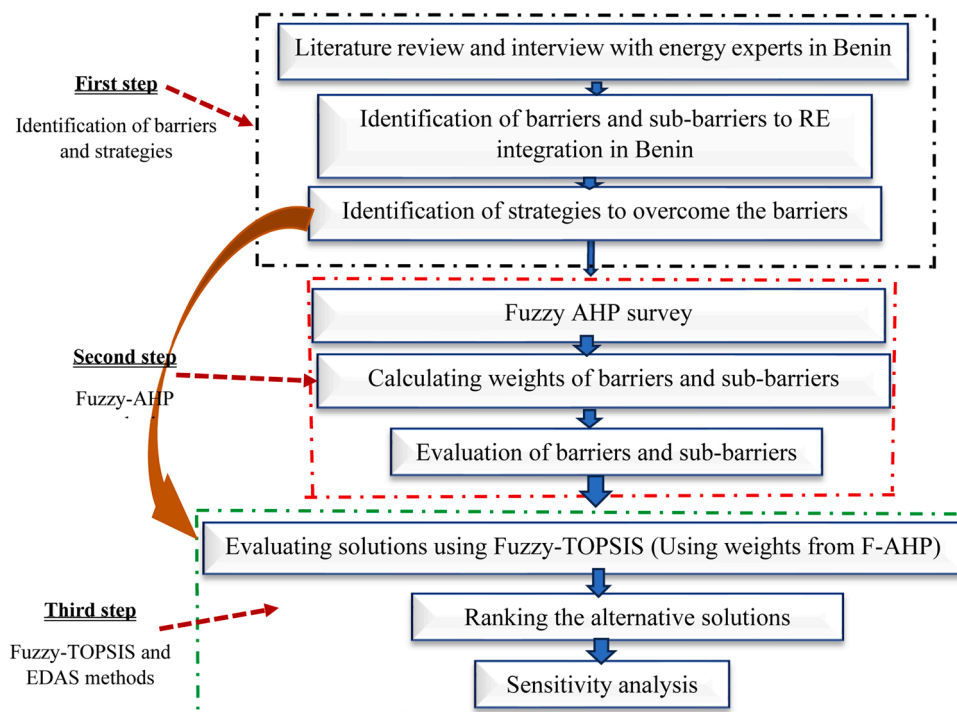


Fig. 3. The general framework of the methodological approach used in the study.

Table 4
The identified barriers to RE development in Benin.

Main barriers	Sub-barriers	Reference
Financial and Economic Barriers (B1)	- Hight Capital Cost (HCC) - Small Market Size (SMS) - Lack of Subsidies and financing mechanism (SFM) - High Operation and Maintenance cost (MOMC)	(Ghimire and Kim, 2018; Sen and Ganguly, 2017; Solangi et al., 2021)
Socio-Cultural Barriers (B2)	- Lack of Information and Public Awareness (IPA) - Lack of Public Acceptance (PA) - Lack of Consumer Paying Capacity (CPC)	(Asante et al., 2022; Gabriel, 2016; Ghimire and Kim, 2018; Gordon et al., 2023; Lee et al., 2022)
Technical Barriers (B3)	- Lack of Local Expertise (LE) - Limited Infrastructure for RE (LI) - Grid Integration Issues (GII) - Lack of Maintenance Facilities (MF) - Intermittency of RE Resources (IRE)	(Dhingra et al., 2022; Gordon et al., 2023; Maman Ali and Yu, 2021; Rahimirad and Sadabadi, 2023; Wyllie et al., 2018)
Political and Regulatory Barriers (B4)	- Lack of Adequate Policy for RE (LAP) - Lack of Institutional collaboration (LIC) - Lack of Governmental Support (GoS)	(Akpahou, Mensah, et al., 2023; Nasirov et al., 2016; Sotiman Yotto et al., 2021; Vanegas Cantarero, 2020)
Geographical and Environmental Barriers (B5)	- Transportation issues (TI) - Land Use and Land Requirement (LUR) - Scattered Households (SH)	(Ghimire and Kim, 2018; Solangi et al., 2021; Vanegas Cantarero, 2020)

formulated by Saaty (Ayhan, 2013). AHP serves as a widely recognized decision-making tool applicable to a diverse range of multi-criteria decision problems. It operates by systematically comparing various alternatives against a set of criteria, offering valuable decision support in the process (Güngör et al., 2009).

In the conventional AHP model, the hierarchy comprises the primary objective at the top level, followed by criteria and sub-criteria in the second and third levels, respectively, with alternatives residing in the fourth level. However, standard AHP cannot accommodate the inherent ambiguity often present in personal judgments (Ayhan, 2013; Dhingra et al., 2022). To address this limitation, the Fuzzy Analytic Hierarchy Process incorporates principles of fuzzy logic. Within the F-AHP framework, both the pairwise comparisons of criteria and alternatives are carried out using linguistic variables, which are effectively represented by triangular numbers (Kilinceci and Onal, 2011; Pan, 2008). This integration of triangular fuzzy numbers into the traditional AHP significantly enhances the decision-makers ability to express nuanced judgments, ultimately leading to improved precision and effectiveness in the decision-making process (Dhingra et al., 2022). As indicated in Fig. 4, $u_{\tilde{A}}(x)=(\ell,m,n)$ is the fuzzy value with

ℓ , m , and n represent the fuzzy numbers associated with the membership function. The fuzzy scale of relative importance used in this study is presented in Table 6.

The F-AHP is described in five main steps in this study as follows:
Step 1: identifying barriers to RE in Benin through literature review and expert interviews. In addition to the literature review, a set of questions was developed and asked experts in the energy sector during interviews to identify the main barriers and strategies to accelerate RE in

Table 5
Proposed strategies to overcome RE barriers in Benin.

Proposed strategies	Description
RE targets and transition plans (S1)	Adequate energy transition plan in Benin will motivate the population and stakeholders to invest in RE technologies. In addition, the introduction of RE targets shall provide a signal and show the importance of RE to public and private actors in the energy sector.
Infrastructure development (S2)	Adequate infrastructure development mechanism is needed to boost the RE deployment in Benin (Akpahou, Mensah, et al., 2023). It is important to invest in RE infrastructure for production, transportation, storage, and distribution for the country's energy sector.
Feed-in tariff (S3)	Currently, there is a lack of effective feed-in tariffs in Benin. However, feed-in tariffs are important to offer long-term agreements and pricing tied to cost for RE producers (Akpahou and Odoi-yorke, 2023; Solangi et al., 2021)
Intergovernmental collaboration (S4)	Collaboration with neighboring countries and international organizations could help mobilize funding and reinforce knowledge-sharing initiatives (Vanegas Cantarero, 2020).
Adequate policy and regulatory framework (S5)	There is a need to develop new clean energy policy and RE regulatory frameworks including permit acquisition and carbon pricing (Solangi et al., 2021). Adequate policy would ensure power quality, and reliability and encourage the adoption of RE by individuals, businesses, and industries (Akpahou and Odoi-yorke, 2023).
Financial assistance and incentives (S6)	Incentives and grant provisions can attract investors and stakeholders to invest in the deployment of RE technologies in Benin (Sotiman Yotto et al., 2021). The government should establish a transparent system to allocate resources and provide grants.
Education and training (S7)	Investing in research and local training programs on RE would help to advance RE technologies and increase local skilled human resources. Research institutions such as Polytechnique College of Abomey-Calavi (EPAC) and the Ecole Supérieure des Métiers des Energies Renouvelables (ESMER) could collaborate with the government to establish regular training programs on RE technologies.
Public awareness campaign (S8)	Awareness campaigns would inform and educate the population about the benefits of the adoption of RE resources. Community awareness including knowledge sharing at schools and colleges would bring more light and information to students and the young population.
Net metering (S9)	The introduction of Net metering in Benin would encourage producers and consumers to invest in RE and reduce distribution and transmission losses (Asante et al., 2022).

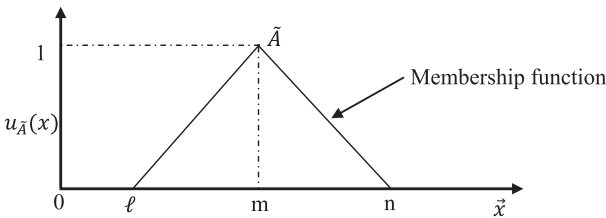


Fig. 4. A triangular fuzzy number (Ali Sadat et al., 2021).

the country. The identified barriers were classified into five main categories and eighteen sub-barriers as presented in Table 4. In addition, nine strategies were proposed to overcome the barriers and are presented in Table 5.

Step 2: a pair-wise comparison matrix is created based on the

Table 6

Fuzzy numbers and language statements of pair-wise comparison (Solangi et al., 2021).

Priority importance	Conventional AHP number	Fuzzy numbers (ℓ, m, n)
Equal	1	(1,1,1)
Moderate	3	(2,3,4)
Strong	5	(4,5,6)
Very strong	7	(6,7,8)
Extremely strong	9	(9,9,9)
Intermediate values	2	(1,2,3)
	4	(3,4,5)
	6	(5,6,7)
	8	(7,8,9)

response from the experts to prioritize and rank the most influential barriers by priority order. The experts were identified from the energy sector of Benin based on their qualifications, the number of years of working experience, and their institutions. Research institutions such as the University of Abomey-Calavi (UAC), governmental organizations, and private institutions were visited to get responses to pre-developed questionnaires which contributed to developing the pair-wise matrix. In addition, the questionnaire was shared on Google form via email and WhatsApp with respondents who could not be physically met. The questionnaire was developed based on the scale of preference with values from 1 to 9 to rate the relative priorities of two criteria (barriers or alternatives) at a time. A criterion is compared to other criteria at a time to determine their priority order. A total of fourteen (14) responses were obtained with 50 % of respondents coming from academia and research institutions, 28.6 % from public agencies, and the remaining 21.4 % coming from private institutions as is presented in Table 7.

Step 3: de-fuzzifying the fuzzy numbers. The fuzzy numbers obtained in the fuzzified pair-wise comparison matrix are de-fuzzified into crisp numbers (Govindan et al., 2015). This is done using the center area method as follows (Eq.3):

$$w_i = \left(\frac{\ell + m + n}{3} \right) \quad (3)$$

Where w_i represents the weight of the criterion i .

Step 4: the global weights of each category of barrier and sub-barriers are estimated and normalized.

3.2.3. Fuzzy TOPSIS technique

The Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) has been employed in many studies in the literature with Table 8 presenting its assessment scale. The fuzzy TOPSIS technique can be presented in the following steps (Ali Sadat et al., 2021;

Table 7

Respondent Characteristics.

	Variable	Obtained records	Percentage
Institution	Research and academia	7	50
	Private agency	3	21.4
	Public agency	4	28.6
Year of experience	Below 5 years	3	21.4
	5–10 years	7	50
	Above 10 years	4	28.6
Job position	Researcher	7	50
	Consultant	3	21.4
	Director	2	14.3
	Project developer	2	14.3
Highest Educational Qualification	Engineer (Ing)	4	28.6
	Master of Science (MSc)	3	21.4
	Doctor of Philosophy (PhD)	7	50

Table 8

Fuzzy TOPSIS technique assesment scale.

Linguistic variables	Fuzzy numbers
Very low	(1,1,3)
Low	(1,3,5)
Average	(3,5,7)
High	(5,7,9)
Very high	(7,9,9)

Chisale and Lee, 2023; Solangi et al., 2021):

Step 1: define the fuzzy decision matrix D using (Eq.4):

$$D = (D_{ab})_{m \times n} \quad (4)$$

Where $D_{ab} = (\ell_{ab}, m_{ab}, n_{ab})$ and $a = 1, 2, 3, \dots, m$ and $b = 1, 2, 3, \dots, n$

Step 2: Determine the normalized fuzzy matrix R as follows (Eq.5):

$$R = (r_{ab})_{m \times n} \quad (5)$$

With

$$r_{ab} = \left(\frac{\ell_{ab}}{n_b^+} + \frac{m_{ab}}{n_b^+} + \frac{n_{ab}}{n_b^+} \right) \quad (6)$$

Where $n_b^+ = \max(n_{ab})$ for beneficial criteria and

$$r_{ab} = \left(\frac{\ell_{ab}^-}{n_{ab}^-} + \frac{\ell_{ab}^-}{m_{ab}^-} + \frac{\ell_{ab}^-}{\ell_{ab}^-} \right) \quad (7)$$

Where $\ell_{ab}^- = \min(\ell_{ab})$ for non-beneficial criterion

Step 3: calculating the weighted normalized fuzzy matrix:

$$V = (v_{ab})_{m \times n} \quad (8)$$

With $v_{ab} = r_{ab} \times w_b$

Step 4: calculating the fuzzy positive and negative solution using (Eq. 9) and (eq.10) respectively:

$$V^+ = (v_1^+, v_2^+, v_3^+, \dots, v_n^+) \quad (9)$$

With $v_b^+ = \max(v_{ab}) \text{ if } (b \in B); \min(v_{ab}) \text{ if } (b \in B')$ (10)

$$V^- = (v_1^-, v_2^-, v_3^-, \dots, v_n^-)$$

With $v_b^- = \max(v_{ab}) \text{ if } (b \in B); \min(v_{ab}) \text{ if } (b \in B')$

Step 5: computing the distance of each alternative from positive and negative fuzzy ideal solution using (Eq. 11) and (eq 12):

$$d_N^+ = \sum_{b=1}^n d(v_{Nb} - v_b^+), \quad N = 1, 2, 3, \dots, m; b = 1, 2, 3, \dots, n \quad (11)$$

$$d_N^- = \sum_{b=1}^n d(v_{Nb} - v_b^-), \quad N = 1, 2, 3, \dots, m; b = 1, 2, 3, \dots, n \quad (12)$$

The distance between two fuzzy numbers $X(x_1, x_2, x_3)$ and $Y(y_1, y_2, y_3)$ is calculated using (Eq.13):

$$d(X, Y) = \sqrt{\frac{1}{3}(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2} \quad (13)$$

Step 6: calculating the closeness coefficient (CC_N) of the alternatives using (Eq. 14):

$$CC_N = \frac{d_N^-}{d_N^+ + d_N^-}, \quad N = 1, 2, 3, \dots, m \quad (14)$$

Step 7: rank and prioritize the alternatives. The alternatives are ranked and prioritized based on the obtained CC_N values.

4. Result and discussion

This section presents and discusses the results obtained for Benin

Republic's long-term energy demand forecasting based on the input data and assumptions made in this study for the various scenarios from 2017 to 2050. It also presents the analysis of prioritizing alternative solutions to RE barriers in the country.

4.1. Energy demand projection analysis

4.1.1. Final energy demand of Benin under the BAU scenario

Fig. 5 presents Benin's final energy demand projection for different activity sectors from 2018 to 2050 under the BAU scenario. It can be seen that the energy demand is expected to increase steadily in each sector, albeit at different growth rates. Specifically, the total energy demand in the country is projected to increase to 248 PJ in 2030 and 445 PJ in 2050, representing a substantial increase of 58 % and 188 %, respectively, compared to the base year energy consumption. The annual average energy consumption growth rate in the BAU scenario is estimated to be around 9 %/year. In 2018, the household sector had the highest contribution to the total energy consumption, representing 44.8 %, followed by the transport sector (43 %), and the commercial services, industry, and agriculture sectors with 10.16 %, 1.6 %, and 0.12 %, respectively. These results are consistent with the share of final energy consumption in Benin reported (PONAME, 2020; SIE-Benin, 2021) for the previous years in Benin.

In addition, the projection shows that the total energy demand of the household and transport sectors could reach 177 PJ and 157 PJ, respectively, in 2050. This finding agrees with the results obtained by Emodi et al. (2017) and Gebremeskel et al., (2021) in Nigeria and Ethiopia respectively, although their estimated annual energy consumption growth rate differs from that of Benin. This difference is expected as countries have different economic and energy situations. This significant increase in energy demand underscores the need for Benin to pursue sustainable energy policies and diversify its energy sources.

4.1.2. Benin Energy demand under the HED scenario

The HED scenario is a projection based on a high economic growth assumption without policy implications built under the BAU scenario. It can be seen in Fig. 6 that the energy demand in the HED scenario is higher than that of the BAU scenario, with a projected total energy demand of 348 PJ and 536 PJ in 2040 and 2050, respectively. These values are approximately 8 % and 21 % higher than the final energy demand in the reference scenario for the same years. This increase in energy demand can be attributed to the assumption of high economic growth, as electricity is a critical input for most economic activities such as agriculture, manufacturing, and transport. As these sectors expand, they require more energy to operate.

Additionally, as the populace becomes wealthy, they use more energy-consuming devices. It is generally acknowledged that there is a positive correlation between a country's energy consumption and its economic growth (Tang et al., 2016). However, the relationship between energy consumption and economic growth is still complex and often misunderstood (Tsafos, 2018). The HED scenarios assume that

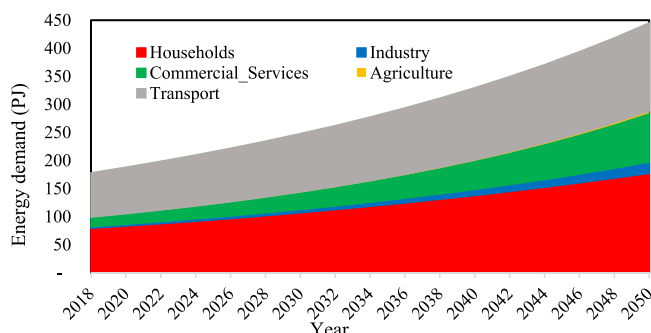


Fig. 5. Final energy demand of Benin under the BAU scenario.

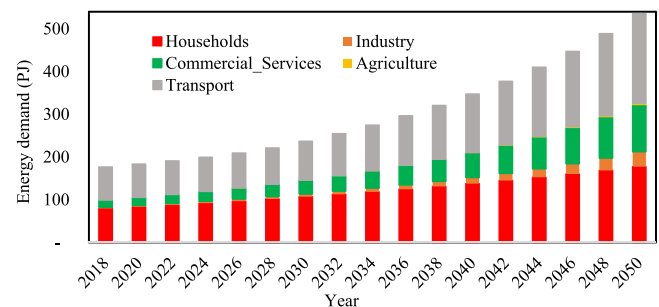


Fig. 6. Final energy demand of Benin under the HED scenario from 2018 to 2050.

there will be strong economic development based on the different projects currently undergoing in Benin, which, as demonstrated, would result in significant energy demand in various sectors. If several energy efficiency measures are not applied to integrate more renewable energy resources, the current supply will not be sufficient to cover future energy demand.

4.1.3. Total energy demand for all scenarios

The analysis shows that energy demand is projected to increase slightly across all scenarios from 2018 to 2050. The HED scenario has the highest energy demand, followed by the BAU, EUG, and HEEA scenarios (Fig. 7). The energy demand under the two policy-driven scenarios is projected to be 360 PJ and 356 PJ in 2050 for EUG and HEEA, respectively. This suggests that low-consuming devices, energy efficiency improvements, and fuel switching decreased energy demand. Notably, the HEEA scenario has higher energy efficiency targets, including switching from fossil fuel-based and traditional cooking stoves.

The projection of final energy consumption by fuel type under the different scenarios from 2018 to 2050 is presented in Fig. 8. Based on the report by INSAE & ICF, (2019), twelve (12) fuel types were considered in this study to capture the complexity of the energy mix in Benin. These comprise electricity, gasoline, kerosene, diesel fuel, residual fuel oil, LPG, coal lignite, firewood, charcoal, vegetable wastes, petroleum coke, and other biomass. Notably, it can be seen that wood has the highest share of combustibles used in Benin, increasing gradually from 68 PJ in 2018–168 PJ in 2050 under the BAU scenario. Firewood alone represents 55.5 % of the total fuel consumption in Benin in 2050, primarily driven by wood used by households mainly for cooking in rural and urban areas.

Furthermore, in Fig. 8 it is shown that wood and gasoline will continue to be most combustible in Benin under the BAU and HED scenarios. The EUG scenario indicates a shift to electricity by 2040. Electricity would take over gasoline consumption in the transportation sector of the country under this scenario. Moreover, in the HEEA scenario, electricity becomes the most consumed combustible with around

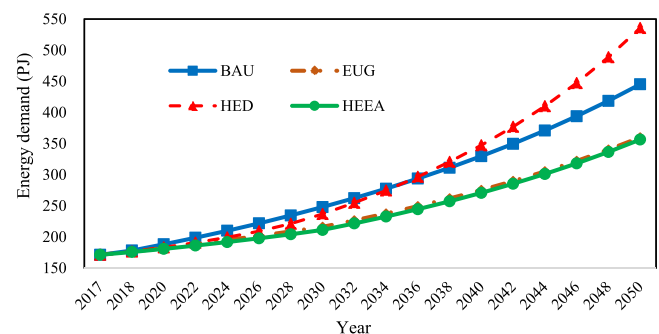


Fig. 7. Total energy demand comparison for all scenarios between 2017 and 2050.

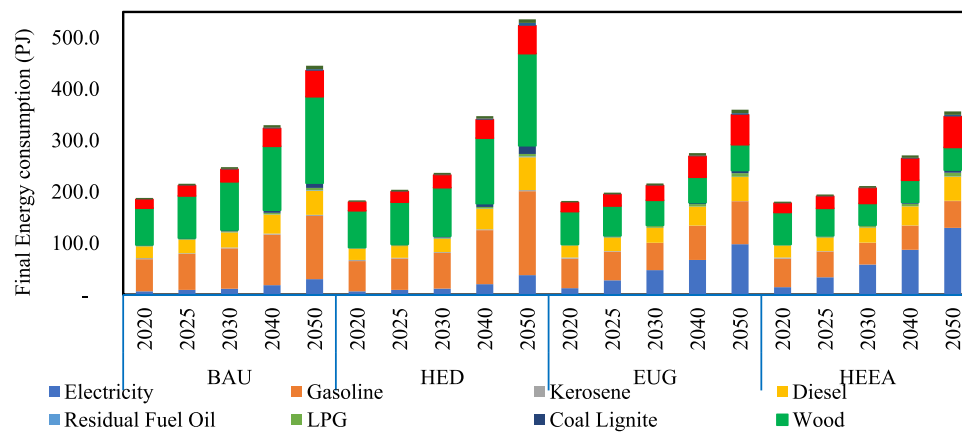


Fig. 8. Total energy by fuel comparison for all scenarios between 2020 and 2050.

130 PJ consumption in 2050. The findings are confirmed by [Plazas-Niño et al., \(2023\)](#), where the electrification of end-uses would dominate the energy transition for developed scenarios in Colombia. In 2050, the energy consumption for HEAA is estimated to be 48.5 % lower than that of the HED scenario. To reduce total energy consumption, the Benin government should prioritize using low-consuming and efficient devices, promote electric cooking stoves, use electric cars, and switch from old and traditional cooking technologies. Such measures align with Sustainable Development Goal 7, emphasizing the need for clean technology adoption. These findings are consistent with results from similar studies conducted in Pakistan ([Raza et al., 2022](#)), Ethiopia ([Gebremeskel et al., 2021](#)), and Nigeria ([Emodi et al., 2017](#)), which have also shown that adequate policy plans and highly efficient energy applications can decrease total energy consumption.

[Ouedraogo \(2017\)](#) projected the final energy demand in West Africa under different scenarios and found that energy consumption can considerably decrease with energy efficiency. The authors estimated the final energy demand of West Africa under the energy efficiency scenario to be 5983 PJ and 6396 PJ in 2030 and 2040, respectively, which are approximately 28 and 23.7 times higher than the figures projected for Benin in this study for the same years. Notably, applying an efficient energy policy positively impacts reducing GHG emissions, further analyzed in the following sub-sections.

4.2. Greenhouse gas (GHG) emissions analysis

4.2.1. GHG emissions under the BAU scenario

[Fig. 9](#) displays the GHG emissions under the BAU scenario projection from 2018 to 2050 in Benin. GHG emissions are increasing over time, reaching a maximal value of 21 million metric tonnes of CO₂ equivalent (MtCO_{2e}) in 2050. Carbon dioxide (CO₂) is the most emitted GHG under

the scenario. It represents 86 % of the total emissions in 2018, while methane (CH₄) and nitrous oxide (NO_x) account for 12.4 % and 1.6 %, respectively. These high CO₂ emissions can be due to the increased use of traditional fuel-based stoves for cooking and the abundant use of gasoline in the transport sector. Gasoline is the most emitting combustible used. It emits approximately half of the total GHG emissions yearly followed by emission from firewood. Notably, gasoline represented more than 73 % of total fuel consumption in the transport sector in 2020 ([SIE-Benin, 2021](#)), considerably increasing the sector's GHG emissions. However, the government of Benin has started implementing actions with laws to tackle deforestation and promote three plantations in the country. Despite that, inappropriate use of non-renewable combustibles for different activities releases considerable CO₂ into the atmosphere ([Bignon et al., 2018](#)), and further actions are required for sustainable development in Benin.

4.2.2. Total GHG emissions for all scenarios

[Fig. 10](#) illustrates the GHG emissions by combustible for the different scenarios while [Table 9](#) shows all scenarios' total GHG emissions in million metric tonnes of CO₂ equivalent. The results indicate that gasoline is the most emitting combustible used in Benin, principally in the transport sector, followed by diesel combustible. The EUG and HEAA scenarios achieved lower GHG emissions than the BAU scenario over time. Specifically, the EUG scenario reduced emissions from 21 MtCO_{2e} in 2050–11.25 MtCO_{2e} compared to BAU, representing a significant emission savings of 9.7 MtCO_{2e}. The HEAA scenario achieved even further emissions reductions due to applying severe energy efficiency targets, resulting in emissions of 9.2 MtCO_{2e} in 2050. In contrast, the no-policy-regulation HED scenario saw an increase in emissions to 26.7 MtCO_{2e}, the highest of all scenarios. These findings highlight the importance of policy-driven measures to curb emissions and promote

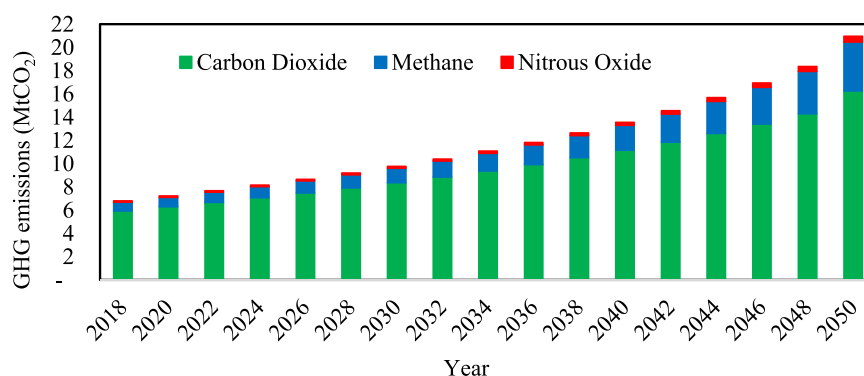


Fig. 9. GHG emissions under BAU scenario from 2018 to 2050.

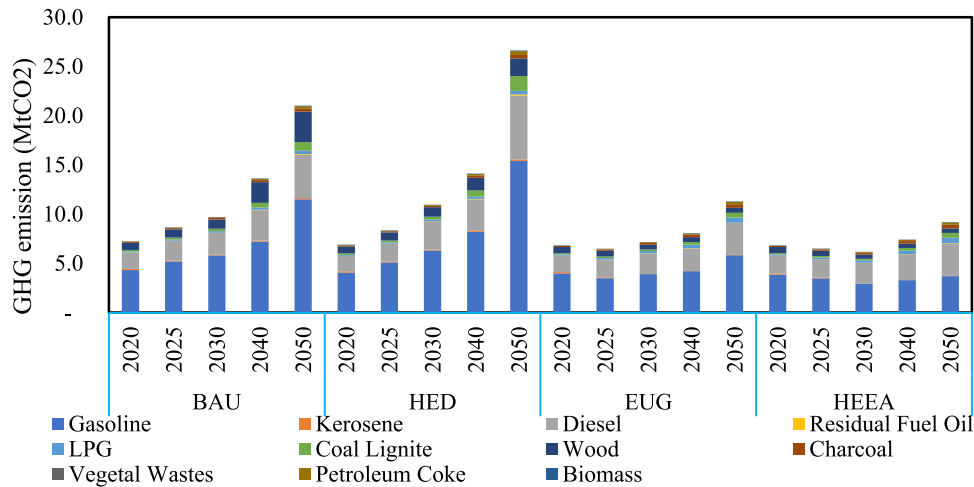


Fig. 10. GHG emissions by combustible comparison for all scenarios between 2020 and 2050.

Table 9

Total GHG emissions for all scenarios (MtCO₂e).

Scenario	2018	2022	2026	2030	2034	2038	2042	2046	2050
BAU	6.8	7.66	8.66	9.77	11.07	12.65	14.33	16.94	20.96
HED	6.8	7.64	8.96	11.07	11.95	13.18	16.22	20.25	26.7
EUG	6.8	6.50	6.85	7.21	7.45	7.54	8.50	9.70	11.25
HEEA	6.8	5.83	5.64	5.36	5.84	6.41	7.11	8.02	9.22

sustainable development. This indicates that in the absence of policy measures to reduce emissions, as energy demand increases, GHG savings decrease. This is a critical finding as it underscores the importance of implementing policies encouraging the transition to low-carbon energy sources and adopting clean energy technologies. The GHG emission reduction analysis further reinforces the need for effective policy measures to support sustainable energy transition. To achieve low-carbon emissions in Benin, deploying clean energy technologies and the use of low-carbon-emitting sources of energy should be promoted through effective policies and regulations.

It is important to mention that the GHG emissions estimated in this study differ from the one in Benin's NDC report (MCVDD, 2021), because of the updated data used from (SIE-Benin, 2021) and the estimation method. Only direct emissions allocated to demand from the energy sector are considered on 100-year Global Warming Potential (GWP). Emissions from transformation and conversion processes are not included. This study employs the latest energy consumption data from the General Directorate of Energy Resources (DGRE) in forecasting energy demand and GHG emissions. The DGRE has now used survey data from the National Institute of Statistics and Demography (INStAD), which did not consider the non-formal importation of petroleum products and mineral coal in the energy supply (SIE-Benin, 2021).

4.3. MCDM analysis results

4.3.1. Barriers to RE integration in Benin

A total of five main barriers and eighteen sub-barriers hindering RE development in Benin prioritized by 14 experts have been evaluated and ranked using the Fuzzy AHP technique. Fig. 11 illustrates the weight distribution among the main barriers (B1 to B5). Notably, the technical barrier (B3) emerges as the most significant obstacle to RE development in Benin, with a weight of 0.304. This is followed by political and regulatory barriers, and financial and economic barriers, ranked second and third, respectively. Table 10 presents the ranking of the main barriers, with B3 being the highest-ranked, followed by B4, B1, B5, and B2, in that order.

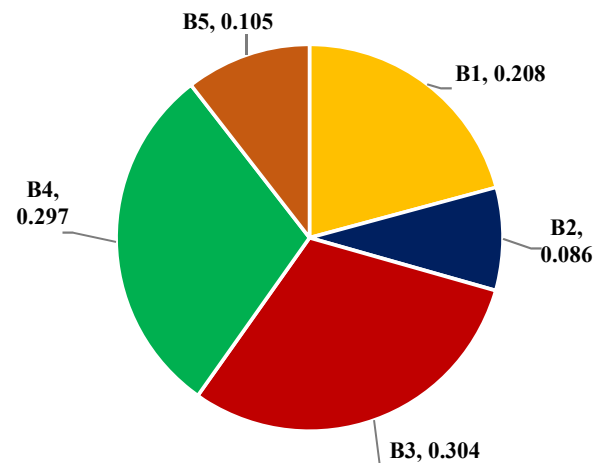


Fig. 11. The weight distribution of the main barriers.

Table 10

The ranking of main barriers.

RE Barriers	Code	Weight	Ranking
Financial and Economic	B1	0.208	3
Socio-Cultural	B2	0.086	5
Technical	B3	0.304	1
Political and Regulatory	B4	0.297	2
Geographical and Environmental	B5	0.105	4

These results suggest that the integration of RE in Benin faces various challenges, with technical barriers taking precedence, followed by political and regulatory, economic and financial, geographical and environmental, and socio-cultural barriers. These results are consistent with similar studies by (Asante et al., 2022) and (Solangi et al., 2021), where technical, political, and financial barriers were also found to be among

the top three hindrances to RE development.

Fig. 12 shows the weights of sub-barriers within each main barrier category. Within the technical barrier category, grid integration issues and limited infrastructure for RE emerge as the most significant sub-barriers, with weights of 0.4148 and 0.3092, respectively. Additionally, high capital cost (HCC) stands out as the principal financial and economic sub-barrier, with a weight of 0.6299. In the political and regulatory barrier category (B4), the lack of institutional collaboration (LIC) is identified as the most significant sub-barrier, representing 68.4 % of all sub-barriers. This is followed by the lack of government support and inadequate RE policy, with weights of 0.1893 and 0.1267, respectively. Transportation issues emerge as the principal sub-barrier within the geographical and environmental barrier category (B5), followed by land use and scattered households.

These findings offer invaluable insights for policymakers, industry stakeholders, and international development agencies seeking to foster sustainable energy transitions in Benin and beyond. Addressing these barriers requires a multifaceted approach, encompassing policy reforms, technological innovation, and strategic investments. Furthermore, the study underscores the urgent need for tailored interventions aimed at overcoming the specific challenges identified within each barrier category. By addressing these barriers in a systematic and coordinated manner, Benin can unlock its vast renewable energy potential, paving the way for a more sustainable and resilient energy future.

The overall ranking of sub-barriers is obtained by multiplying the weight of each sub-barrier by the priority weights of each main barrier. The overall weights and ranking of sub-barriers are presented in Table 11. The findings highlight that the lack of institutional collaboration, high capital cost, grid integration issues, limited infrastructure, and transportation issues emerge as the top five critical barriers hindering Benin’s sustainable energy transition. Notably, the first two barriers belong to the financial and economic, and political and regulatory barrier categories, respectively. In light of these results, it becomes evident that for Benin to expedite the transition to clean energy and elevate the share of renewable energy in the country’s final energy supply mix, addressing economic and political barriers should be given top priority. This underscores the importance of government intervention and policy reforms in overcoming key obstacles to sustainable energy adoption.

Interestingly, geographical and environmental barriers are ranked as the least concerning obstacles. This suggests that while geographical and environmental factors may present challenges, they are not perceived as significant barriers compared to economic and political factors. These findings resonate with similar studies conducted in Pakistan (Solangi et al., 2021) and Malawi (Chisale and Lee, 2023) where economic and

Table 11
Overall weights and ranking of sub-barriers.

RE Sub-Barriers	Code	Weight	Ranking
Hight Capital Cost	HCC	0.1310	2
Small Market Size	SMS	0.0331	11
Lack of Subsidies and financing mechanism	SFM	0.0361	10
High Operation and Maintenance cost	MOMC	0.0078	18
Lack of Information and Public Awareness	IPA	0.0369	9
Lack of Public Acceptance	PA	0.0279	12
Lack of Consumer Paying Capacity	CPC	0.0211	14
Lack of Local Expertise	LE	0.0202	15
Limited Infrastructure for RE	LI	0.0940	4
Grid Integration Issues	GII	0.1261	3
Lack of Maintenance Facilities	MF	0.0216	13
Intermittency of RE Resources	IRE	0.0422	7
Lack of Adequate Policy for RE	LAP	0.0376	8
Lack of Institutional collaboration	LIC	0.2031	1
Lack of Governmental Support	GoS	0.0562	6
Transportation issues	TI	0.0718	5
Land Use and Land Requirement	LUR	0.0132	17
Scattered Households	SH	0.0200	16

political barriers were identified as dominant, while geographical and environmental barriers were ranked lower. This consistency across different regions underscores the universal challenges facing sustainable energy transitions and underscores the importance of addressing economic and political barriers on a global scale.

4.3.2. Strategies to promote RE integration in Benin

The Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) methodology is utilized to assess and prioritize alternative solutions aimed at addressing the identified barriers within the Republic of Benin. Table 12 shows the weights and the ranking of the alternatives based on their closeness coefficient (CC_N). Among the

Table 12
Weights and ranking of the proposed alternatives.

Proposed strategies	d_N^+	d_N^-	CC _N	Rank
S1	0.0840	0.1331	0.6132	2
S2	0.0933	0.1237	0.5701	4
S3	0.1625	0.0545	0.2513	9
S4	0.1056	0.1115	0.5136	5
S5	0.0819	0.1352	0.6227	1
S6	0.1426	0.0745	0.3432	8
S7	0.0929	0.1241	0.5719	3
S8	0.1397	0.0773	0.3562	7
S9	0.1172	0.0999	0.4601	6

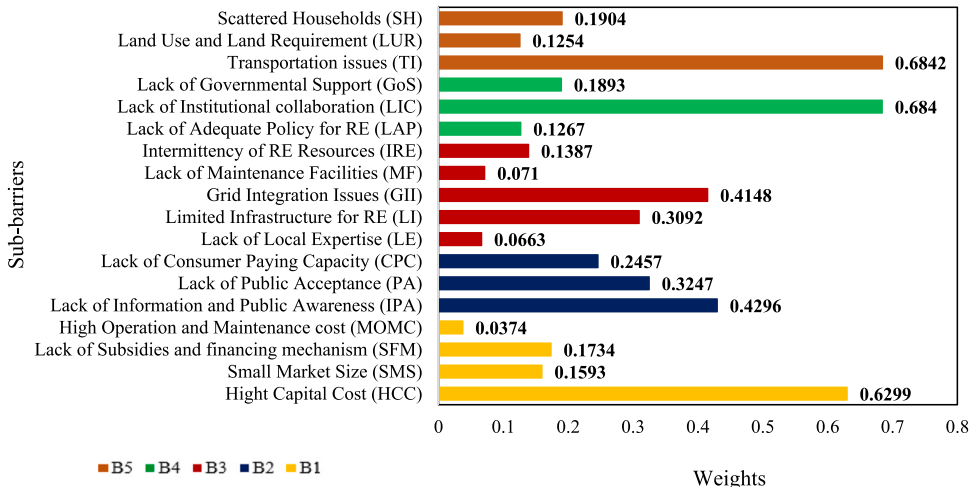


Fig. 12. Weights of sub-barriers within each main barrier category.

alternatives, the implementation of an adequate policy and regulatory framework (S5) emerges as the most effective solution for eliminating barriers to Renewable Energy (RE) deployment in the country, with a substantial weight of 0.6227. Following closely are alternative solutions such as RE targets and transition plans (S1), education and training (S4), and infrastructure development (S2), ranked as the second, third, and fourth options, respectively. It's important to note that establishing robust policy and regulatory frameworks provides the foundation for fostering RE development by creating conducive environments for investment and innovation. Additionally, setting clear RE targets and transition plans, investing in education and training initiatives, and enhancing infrastructure development are crucial steps in overcoming the identified barriers and accelerating the transition towards sustainable energy in Benin.

4.3.3. Sensitivity analysis

A sensitivity analysis was performed to validate the findings of the study. This analysis aimed to explore how the ranking of sub-barriers and proposed alternative solutions change when the weights assigned to the main barriers are varied. Six different scenarios were examined by adjusting the weights of the barriers, as outlined in Table 13. Case I allocated equal weights to all main barriers, while the weights varied in the other scenarios. The rankings of sub-barriers resulting from the sensitivity analysis are depicted in Fig. 13. Despite fluctuations in rankings across different scenarios, high capital costs (HCC), lack of institutional collaboration (LIC), and grid integration issues (GII) consistently emerged as the top barriers in most cases. Furthermore, the sensitivity analysis conducted on the proposed alternatives, shown in Fig. 14, revealed that solutions S1, S2, and S7 remained among the top 5 alternatives deemed essential for addressing barriers and expediting the transition towards sustainable energy.

4.4. Policy implications

As suggested by the EUG and HEEA scenarios, the government of Benin should prioritize promoting modern and low-carbon emitting devices to achieve sustainable development. To ensure universal access to energy for all Beninese by 2030, integrating more renewable resources and an adequate energy policy is crucial to reducing environmental emissions. Benin's National Adaptation Program of Action has set goals to help local communities adapt to the effects of climate change, including promoting renewable energy and low-emission home technologies in vulnerable regions (UNDP, 2023). Implementing effective programs and policies is vital to addressing the country's electricity problems and achieving SDG 7 (Akpahou et al., 2024).

Moreover, promoting renewable energy sources, such as solar, wind, and hydropower, as demonstrated by the results is critical to meet the increasing energy demand while mitigating fuel-based resources' adverse effects. The development of bioenergy projects could contribute to the valorization of biomass resources. Benin's waste management company (SGDS) collects huge amounts of waste in the 5 cities of the Grand-Nokoué, which are buried in the ground or burnt in the open air it would be crucial to develop technologies to convert the waste into energy for recycling. In addition, the effective development of adequate energy policies will enable the government of Benin to boost investment

in the power sector while creating job opportunities for the local population. Encouraging energy efficiency and optimization measures in the industrial and commercial sectors will also reduce energy consumption and increase energy productivity. Furthermore, international collaboration and support will be necessary to help Benin achieve its sustainable energy goals and promote economic development.

Investing in the expansion and modernization of the energy grid is crucial for integrating renewable energy sources into the existing electricity grid. Renewable energy generation can be effectively integrated by improving grid capacity and reliability, ensuring a reliable and stable energy supply. Additionally, grid expansion efforts can enhance energy access in remote areas, extending the benefits of modern energy services to underserved communities. Supporting research and development efforts in renewable energy technologies could also accelerate their adoption and make them more cost-competitive. Developing adequate policy and regulatory framework (S5), RE targets and transition plans (S1), education and training (S4), and infrastructure development (S2), as suggested by the study results, would accelerate clean energy transition. This could also lead to the development of innovative solutions to improve the efficiency, affordability, and scalability of RE technologies, fostering technological advancements and driving the transition towards a cleaner and more sustainable energy future.

4.5. Limitations and future work

The modeling approach employed in this study entails certain limitations that necessitate acknowledgment and identification of areas for future research. A primary challenge arises from the limited availability of data on energy consumption and demand patterns within different sectors in Benin. As this study is the first of its kind to be conducted in Benin, it used data from neighboring developing countries with similar patterns and global trends in West Africa. The assumptions made are based on data collected from secondary sources and governmental reports in Benin. The assumptions were conservative as much as possible, however, there is still inherent uncertainty in future energy demand and room for further improvement. Also, the study does not include a cost analysis of the proposed alternative policies in Benin. Future research could include a cost analysis of the scenarios developed in this study and their impacts on the energy system of the country. Moreover, a critical area for exploration involves determining the optimal renewable energy supply mix required to meet Benin's energy demand and facilitate a seamless transition to clean energy. A more detailed modeling of renewable energy sources, including solar PV, wind energy, hydropower, and bioenergy abundantly available in Benin has the potential to significantly enrich and complement the findings of this study. Expanding the scope of the investigation, future research could delve into the analysis of energy generation in a supply scenario specifically tailored for the transformation sector of Benin. Additionally, conducting a techno-economic analysis of energy efficiency and renewable energy policy development presents a promising avenue for further exploration.

5. Conclusion

In this study, an energy model is developed for the Benin Republic, considering important factors that could influence future energy plans in the country. Four scenarios examined alternative pathways development for Benin's energy sector between 2017 and 2050. The four scenarios developed include a reference scenario (BAU) and three alternative scenarios (HED, EUG, and HEEA), considering various levels of policy and GHG emissions reduction targets. Fuzzy AHP and TOPSIS techniques have been employed to prioritize alternative solutions to RE barriers in the country. The main findings are as follows:

Under the BAU scenario, energy demand is projected to increase with a growth rate of around 9 % per year, reaching the total energy demand of 445 PJ in 2050. This represents an increase of 188 % compared to the energy demand in 2017. However, with no policy action and a high

Table 13
Main barriers' weight for sensitivity analysis.

Barriers	Current Weights	Case I	Case II	Case III	Case IV	Case V	Case VI
B1	0.208	0.2	0.35	0.125	0.125	0.125	0.275
B2	0.086	0.2	0.275	0.35	0.125	0.125	0.125
B3	0.304	0.2	0.125	0.275	0.35	0.125	0.125
B4	0.297	0.2	0.125	0.125	0.275	0.35	0.125
B5	0.105	0.2	0.125	0.125	0.125	0.275	0.35
Total	1	1	1	1	1	1	1

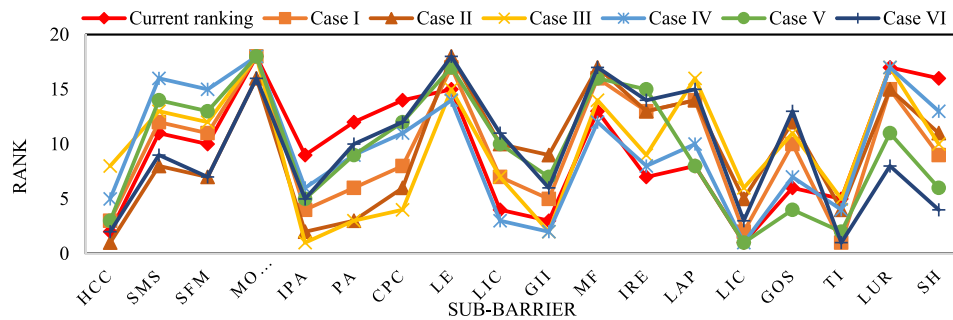


Fig. 13. Sub-barrier ranking for the sensitivity analysis.

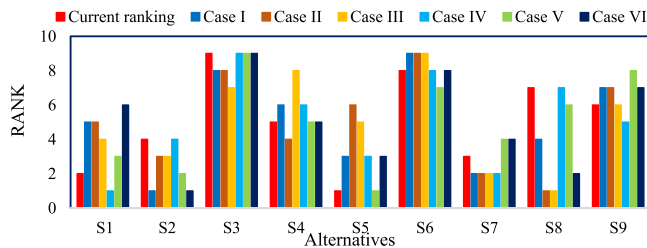


Fig. 14. Alternative solutions ranking for the sensitivity analysis.

economic development situation (HED scenario), the energy demand would increase by 20 % higher than the final energy demand of the BAU scenario in 2050. Household and transport are the most energy-demanding sectors. Notably, the energy demand under the two policy-driven scenarios has become much lower and is projected to be 360 PJ and 356 PJ for EUG and HEEA scenarios, respectively in 2050. The total energy demand projected under the HEEA scenario is 48.5 % lower than that of the HED scenario in 2050. This suggests that applying adequate policy and willingness to reduce carbon emissions could help significantly decrease energy consumption in Benin. Furthermore, the GHG emissions analysis indicates that the EUG and HEEA scenarios achieved lower emissions compared to the BAU and HED scenarios. The GHG emissions are projected to amount to 26.7 MtCO_{2e}, 21 MtCO_{2e}, 11.25 MtCO_{2e}, and 9.2 MtCO_{2e} for HED, BAU, EUG, and HEEA scenarios, respectively, in 2050. These results show the importance of policy-driven measures to curb emissions and promote sustainable development.

The analysis conducted through the Fuzzy AHP reveals the primary obstacles hindering the deployment of RE in Benin, prioritized as follows: Technical> Political and Regulatory>Financial and Economic>Geographical and Environmental> Socio-Cultural. Notably, the government should concentrate its efforts on addressing technical, political, and financial barriers as they rank as the most significant impediments to expediting the transition towards clean energy. Among the sub-barriers identified, lack of institutional collaboration emerges as the foremost concern, followed by high capital costs, grid integration issues, limited infrastructure, and transportation challenges. Moreover, employing the Fuzzy TOPSIS method to rank potential solutions underscores the importance of establishing a robust policy and regulatory framework (S5) as the paramount strategy to mitigate RE barriers in Benin. Following closely are initiatives such as setting RE targets and transition plans (S1), investing in education and training (S4), and enhancing infrastructure development (S2), recognized as the second, third, and fourth most effective alternatives, respectively.

It is crucial for the government of Benin to prioritize the use of low-consuming and efficient devices and promote the use of electric cars and improved cooking stoves to reduce total energy consumption. Important policy measures are needed to enable the energy transition and reduce GHG emissions. Even though implementing new energy policies is

costly, they are critical in achieving SDG 7. International collaboration and financial support are required for sustainable energy and economic development. The country possesses a rich and diverse RE potential that could significantly contribute to the country's sustainable development and energy security.

CRediT authorship contribution statement

Romain Akpahou: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Lena D Mensah:** Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization. **David A. Quansah:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Francis Kemausuor:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgment

Romain Akpahou acknowledges the Fostering Research & Intra-African knowledge transfer through Mobility & Education (FRAME) scholarship program funded by the European Union (project number: 2019-1973/007-001), for providing him with a scholarship to pursue a Ph.D. program at Kwame Nkrumah University of Science and Technology (KNUST).

References

- Adamon, G.F., D. Jossou, A., A. Adomou, N., A. Pinta, F., 2020. Evaluation of the energy potential of agricultural waste in West Africa from three biomasses of interest in Benin. *Int. J. Adv. Res.* 8 (4), 766–774. <https://doi.org/10.21474/ijar01/10839>.
- Adeyemi-Kayode, T., Misra, S., Orovwode, H., Adoghe, A., 2022. Modeling the next decade of energy sustainability: a case of a developing country. *Energies* 15 (14), 1–19. <https://doi.org/10.3390/en15145083>.
- Akpahou, R., Mensah, L.D., Quansah, D.A., 2023. Renewable energy in Benin: current situation and future prospects. *Clean Energy* 7 (5), 952–961.
- Akpahou, R., Odoi-yorke, F., 2023. A multicriteria decision-making approach for prioritizing renewable energy resources for sustainable electricity generation in Benin. *Cogent Eng.* 10 (1) <https://doi.org/10.1080/23311916.2023.2204553>.
- Akpahou, R., Odoi-yorke, F., Kwasi, L., 2023. Techno-economic analysis of a utility-scale grid-tied solar photovoltaic system in Benin republic. *Clean. Eng. Technol.* 13 (April), 100633 <https://doi.org/10.1016/j.clet.2023.100633>.
- Akpahou, R., Odoi-Yorke, F., Mensah, L.D., Quansah, D.A., Kemausuor, F., 2024. Strategizing towards sustainable energy planning: modeling the mix of future generation technologies for 2050 in Benin. *Renew. Sustain. Energy Transit.* 5 (January), 100079 <https://doi.org/10.1016/j.rset.2024.100079>.

- Ali Sadat, S., Vakilaroaya Fini, M., Hashemi-Dezaki, H., Nazififard, M., 2021. Barrier analysis of solar PV energy development in the context of Iran using fuzzy AHP-TOPSIS method. *Sustain. Energy Technol. Assess.* 47 (August), 101549. <https://doi.org/10.1016/j.seta.2021.101549>.
- Asante, D., Dankwa, J., Afrane, S., Adjei-darko, P., Asante, B., Fosu, E., Ampah, D., Oppong, P., 2022. Prioritizing strategies to eliminate barriers to renewable energy adoption and development in Ghana: a CRITIC-fuzzy TOPSIS approach. *Renew. Energy* 195, 47–65. <https://doi.org/10.1016/j.renene.2022.06.040>.
- Ayhan, M.B., 2013. A fuzzy Ahp approach for supplier selection problem: a case study in a gearmotor company. *Int. J. Manag. Value Supply Chains* 4 (3), 11–23. <https://doi.org/10.5121/ijmvsc.2013.4302>.
- Bignon, D., Houndjo, M., Adjolohoun, S., Gbenou, B., Saidou, A., Ahoton, L., Houninato, M., Toleba, S.S., Sinsin, B.A., 2018. Socio-Demogr. Econ. Charact., Crop-Livest. Prod. Syst. Issues rearing Improv.: A Rev. 12 (February), 519–541.
- Çelikbilek, Y., Tüysüz, F., 2016. An integrated grey based multi-criteria decision making approach for the evaluation of renewable energy sources. *Energy* 115, 1246–1258. <https://doi.org/10.1016/j.energy.2016.09.091>.
- Chisale, S.W., Lee, H.S., 2023. Evaluation of barriers and solutions to renewable energy acceleration in Malawi, Africa, using AHP and fuzzy TOPSIS approach. *Energy Sustain. Dev.* 76 (June), 101272. <https://doi.org/10.1016/j.esd.2023.101272>.
- Dhingra, T., Sengar, A., Sajith, S., 2022. A fuzzy analytic hierarchy process-based analysis for prioritization of barriers to offshore wind energy. *J. Clean. Prod.* 345 (July 2021), 131111. <https://doi.org/10.1016/j.jclepro.2022.131111>.
- Emodi, N.V., Emodi, C.C., Murthy, G.P., Emodi, A.S.A., 2017. Energy policy for low carbon development in Nigeria: A LEAP model application. *Renew. Sustain. Energy Rev.* 68 (August 2015), 247–261. <https://doi.org/10.1016/j.rser.2016.09.118>.
- Energy Commission Ghana. (2020). National energy statistics 2000 - 2019, strategic planning and policy directorate (Issue April). (<https://doi.org/10.1088/0031-9112/27/9/008>).
- Gabriel, C.A., 2016. What is challenging renewable energy entrepreneurs in developing countries? *Renew. Sustain. Energy Rev.* 64, 362–371. <https://doi.org/10.1016/j.rser.2016.06.025>.
- GDIZ. (2023). Implementation of the Glo-Djigbé Special Economic Zone project| Government of the Republic of Benin. (French). (<https://www.gouv.bj/article/1163/mise-oeuvre-projet-zone-economique-speciale-djigbe-300.000-emplois-annonces/>).
- Gebremeskel, D.H., Ahlgren, E.O., Beyene, G.B., 2021. Long-term evolution of energy and electricity demand forecasting: the case of Ethiopia. *Energy Strategy Rev.* 36 (August 2020), 100671. <https://doi.org/10.1016/j.esr.2021.100671>.
- Gebremeskel, D.H., Ahlgren, E.O., Beyene, G.B., 2023. Long-term electricity supply modelling in the context of developing countries: the OSeMOSYS-LEAP soft-linking approach for Ethiopia. *Energy Strategy Rev.* 45 (June 2022), 101045. <https://doi.org/10.1016/j.esr.2022.101045>.
- Ghimire, L.P., Kim, Y., 2018. An analysis on barriers to renewable energy development in the context of Nepal using AHP. *Renew. Energy* 129, 446–456. <https://doi.org/10.1016/j.renene.2018.06.011>.
- Global data. (2023). Average household size - Area Database - Table - Global Data Lab (Accessed 2023-03-13). (<https://globaldatalab.org/areadata/table/hsize/COD/>).
- Gordon, J.A., Balta-Ozkan, N., Nabavi, S.A., 2023. Socio-technical barriers to domestic hydrogen futures: repurposing pipelines, policies, and public perceptions. *Appl. Energy* 336 (February), 120850. <https://doi.org/10.1016/j.apenergy.2023.120850>.
- Govindan, K., Diabat, A., Madan Shankar, K., 2015. Analyzing the drivers of green manufacturing with fuzzy approach. *J. Clean. Prod.* 96, 182–193. <https://doi.org/10.1016/j.jclepro.2014.02.054>.
- Güngör, Z., Serhadlioglu, G., Kesen, S.E., 2009. A fuzzy AHP approach to personnel selection problem. *Appl. Soft Comput.* 9 (2), 641–646. <https://doi.org/10.1016/j.asoc.2008.09.003>.
- Huang, Y., Bor, Y.J., Peng, C.Y., 2011. The long-term forecast of Taiwan's energy supply and demand: LEAP model application. *Energy Policy* 39 (11), 6790–6803. <https://doi.org/10.1016/j.enpol.2010.10.023>.
- Hussain, A., Perwez, U., Ullah, K., Kim, C.H., Asghar, N., 2021. Long-term scenario pathways to assess the potential of best available technologies and cost reduction of avoided carbon emissions in an existing 100% renewable regional power system: a case study of Gilgit-Baltistan (GB), Pakistan. *Energy* 221, 119855. <https://doi.org/10.1016/j.energy.2021.119855>.
- IEA, IRENA, UNSD, W. B. WHO, 2021. Tracking SDG7: The Energy Progress Report.
- INSAE, & I.C.F. (2019). National Institute of Statistics and Economic Analysis. Demographic and Health Survey in Benin, 2017–2018. Cotonou, Benin and Rockville, Maryland, USA: INSAE and ICF. (French).
- INStad. (2022). Population projections from 2014 to 2063 and outlook for social demand from 2014 to 2030 in Benin. (French).
- IRENA. (2022). *Country Indicators and Sdgs: Benin*. Masdar City, Abu Dhabi. (https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Africa/Benin_Africa_RE.SP.pdf) (Accessed November 30, 2022).
- Kilinc, O., Onal, S.A., 2011. Fuzzy AHP approach for supplier selection in a washing machine company. *Expert Syst. Appl.* 38 (8), 9656–9664. <https://doi.org/10.1016/j.eswa.2011.01.159>.
- Lee, Y., Lee, M.C., Kim, Y.J., 2022. Barriers and strategies of hydrogen fuel cell power generation based on expert survey in South Korea. *Int. J. Hydrog. Energy* 47 (9), 5709–5719. <https://doi.org/10.1016/j.ijhydene.2021.11.212>.
- Machado, P.G., Mouette, D., Villanueva, L.D., Esparta, A.R., Mendes Leite, B., Moutinho dos Santos, E., 2019. Energy systems modeling: trends in research publication. *Wiley Interdiscip. Rev. Energy Environ.* 8 (4), 1–15. <https://doi.org/10.1002/wene.333>.
- Mahboob, M., Ali, M., ur Rashid, T., Hassan, R., 2021. Pakistan's energy demand forecasting for various sector through long range alternative planning system. *Eng. Proc.* 12 (1), 10–13. <https://doi.org/10.3390/engproc2021012020>.
- Maman Ali, M.M., Yu, Q., 2021. Assessment of the impact of renewable energy policy on sustainable energy for all in West Africa. *Renew. Energy* 180, 544–551. <https://doi.org/10.1016/j.renene.2021.08.084>.
- MCVDD. (2021). Nationally Determined Contribution of Benin under the Paris Agreement. Updated version (Final document). (French).
- Mekonnen, T.W., Teferi, S.T., Kebede, F.S., Anandarahaj, G., 2022. Assessment of Impacts of climate change on hydropower-dominated power system—the case of Ethiopia. *Appl. Sci. (Switz.)* 12 (4). <https://doi.org/10.3390/app12041954>.
- Mensah, J.H.R., dos Santos, I.F.S., Tiago Filho, G.L., 2022. A critical analysis of the energy situation in the Benin republic and its evolution in last decade. *SSRN Electron. J.* 55 (35). <https://doi.org/10.2139/ssrn.4074515>.
- Mensah, J.H.R., dos Santos, I.F.S., Tiago Filho, G.L., 2023. A critical analysis of the energy situation in the Benin Republic and its evolution over the last decade. *Renew. Energy* 202 (October 2022), 634–650. <https://doi.org/10.1016/j.renene.2022.11.085>.
- Ministry of Energy. (2020). National Renewable Energy Development Policy (Ponader). (French). (<https://energie.gouv.bj/article/politique-nationale-de-developpement-des-energies-renouvelables>).
- Mondal, M.A.H., Boie, W., Denich, M., 2010. Future demand scenarios of Bangladesh power sector. *Energy Policy* 38 (11), 7416–7426. <https://doi.org/10.1016/j.enpol.2010.08.017>.
- Nasirov, S., Silva, C., Agostini, C.A., 2016. Assessment of barriers and opportunities for renewable energy development in Chile. *Energy Sources, Part B: Econ., Plan. Policy* 11 (2), 150–156. <https://doi.org/10.1080/15567249.2015.1062820>.
- Neniskis, E., Galinis, A., Norvaisa, E., 2021. Improving transport modeling in message energy planning model: Vehicle age distributions. *Energies* 14 (21). <https://doi.org/10.3390/en14212729>.
- Numata, M., Sugiyama, M., Mogi, G., 2020. Barrier analysis for the deployment of renewable-based mini-grids in Myanmar using the analytic hierarchy process (AHP). *Energies* 16 (3). <https://doi.org/10.3390/en13061400>.
- Odou, O.D.T., Bhandari, R., Adamou, R., 2020. Hybrid off-grid renewable power system for sustainable rural electrification in Benin. *Renew. Energy* 145, 1266–1279. <https://doi.org/10.1016/j.renene.2019.06.032>.
- Ouedraogo, N.S., 2017. Africa energy future: Alternative scenarios and their implications for sustainable development strategies. *Energy Policy* 106 (March), 457–471. <https://doi.org/10.1016/j.enpol.2017.03.021>.
- Pan, N.F., 2008. Fuzzy AHP approach for selecting the suitable bridge construction method. *Autom. Constr.* 17 (8), 958–965. <https://doi.org/10.1016/j.autcon.2008.03.005>.
- Perwez, U., Sohail, A., Hassan, S.F., Zia, U., 2015. The long-term forecast of Pakistan's electricity supply and demand: An application of long range energy alternatives planning. *Energy* 93 (Part 2), 2423–2435. <https://doi.org/10.1016/j.energy.2015.10.103>.
- Plazas-Niño, F.A., Yeganyan, R., Cannone, C., Howells, M., Quirós-Tortós, J., 2023. Informing sustainable energy policy in developing countries: An assessment of decarbonization pathways in Colombia using open energy system optimization modelling. *Energy Strategy Rev.* 50 (October). <https://doi.org/10.1016/j.esr.2023.101226>.
- PONAME. (2020). National policy of Energy management (POLITIQUE NATIONALE DE MAITRISE D'ENERGIE 2020-2030). (https://pspbd.dev.gouv.bj/server/storage/app/PolitiqueFichiers/63_PONAME_VF.pdf) (Accessed November 30, 2022).
- Rahimirad, Z., Sadabadi, A.A., 2023. Green hydrogen technology development and usage policymaking in Iran using SWOT analysis and MCDM methods. *Int. J. Hydrog. Energy* 48 (40), 15179–15194. <https://doi.org/10.1016/j.ijhydene.2023.01.035>.
- Raza, M.A., Khatri, K.L., Israr, A., Ul Haque, M.I., Ahmed, M., Rafique, K., Saand, A.S., 2022. Energy demand and production forecasting in Pakistan. *Energy Strategy Rev.* 39, 100788. <https://doi.org/10.1016/j.esr.2021.100788>.
- Sam-Amobi, C., Ekechukwu, O.V., Chukwualu, C.B., 2019. A preliminary assessment of the energy related carbon emissions associated with hotels in Enugu Metropolis Nigeria. *AFRREV STECH: Int. J. Sci. Technol.* 8 (2), 19–30. <https://doi.org/10.4314/stech.v8i2.2>.
- SE4ALL. (2023). Benin - SEforALL Africa Hub (accessed, 2023-03-14). (<https://www.se4all-africa.org/seforall-in-africa/country-data/benin/>).
- SEI, 2005. *Stockh. Environ. Institut. Use Guide LEAP 2005*. May 2005.
- Sen, S., Ganguly, S., 2017. Opportunities, barriers and issues with renewable energy development – A discussion. *Renew. Sustain. Energy Rev.* 69 (June 2015), 1170–1181. <https://doi.org/10.1016/j.rser.2016.09.137>.
- Shahzad, K., Lu, B., Abdul, D., 2022. Entrepreneur barrier analysis on renewable energy promotion in the context of Pakistan using Pythagorean fuzzy AHP method. *Environ. Sci. Pollut. Res.* 54756–54768. <https://doi.org/10.1007/s11356-022-19680-3>.
- SIE-Benin. (2021). Key Figures 2021, Energy Balances and Indicators 2016 to 2020. (French). 1–26. (<https://direction-energie.gouv.bj/actualites/sie-benin-atelier-national-de-presentation-du-systeme-dinformation-energetique-du-benin-3.html>) (Accessed November 23, 2022).
- Solangi, Y.A., Longsheng, C., Shah, S.A.A., 2021. Assessing and overcoming the renewable energy barriers for sustainable development in Pakistan: An integrated AHP and fuzzy TOPSIS approach. *Renew. Energy* 173, 209–222. <https://doi.org/10.1016/j.renene.2021.03.141>.
- Sotiman Yotto, H.C., Chetangny, P.K., Aredjodoun, J., Houndedako, S., Chamagne, D., Barbier, G., Vianou, A., 2021. The renewable energy landscape in Benin: An analysis and review of barriers, targets, policies and actions for a clean energy transition. *Int. Conf. Electr., Comput., Commun. Mechatron. Eng., ICECCME 2021*, Oct., 7–8. <https://doi.org/10.1109/ICECCME52200.2021.9590886>.
- Tang, C.F., Tan, B.W., Ozturk, I., 2016. Energy consumption and economic growth in Vietnam. *Renew. Sustain. Energy Rev.* 54, 1506–1514. <https://doi.org/10.1016/j.rser.2015.10.083>.

- Tsafos, N. (2018). *Energy and Growth: Exploring a Nuanced Relationship*. (<https://doi.org/10.1787/DATA-00514-EN>).
- UNDP. (2023). *Benin National Adaptation Programme of Action (NAPA) | UNDP Project Portal* (Accessed 2023-03-27). (<https://www.gcfprojects-undp.org/projects/benin-national-adaptation-programme-action-napa>).
- United Nations. (2018). *Lead Organizations With the financial support from Governments of Norway, Netherlands, and China through the UN sub-trust fund for the 2030 Agenda for Sustainable Development as well as the European Commission, ENERGIA and HIVOS*. (https://sustainabledevelopment.un.org/content/documents/18041SDG7_Policy_Brief.pdf).
- Vanegas Cantarero, M.M., 2020. Of renewable energy, energy democracy, and sustainable development: A roadmap to accelerate the energy transition in developing countries. *Energy Res. Soc. Sci.* 70 (November 2019), 101716 <https://doi.org/10.1016/j.erss.2020.101716>.
- Wambui, V., Njoka, F., Muguthu, J., Ndwalu, P., 2022. Scenario analysis of electricity pathways in Kenya using Low Emissions Analysis Platform and the Next Energy Modeling system for optimization. *Renew. Sustain. Energy Rev.* 168 (August), 112871 <https://doi.org/10.1016/j.rser.2022.112871>.
- World bank. (2023). Access to electricity (% of population) - Benin | Data (accessed 2023-03-14). (<https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=BJ>).
- World population review. (2023). Benin Population 2022 (Demographics, Maps, Graphs), accessed March, 2023. (<https://worldpopulationreview.com/countries/benin-population>).
- Wyllie, J.O.Y., Essah, E.A., Ofetotse, E.L., 2018. Barriers of solar energy uptake and the potential for mitigation solutions in Barbados. *Renew. Sustain. Energy Rev.* 91 (March), 935–949. <https://doi.org/10.1016/j.rser.2018.04.100>.
- Yatokpa, A.B. (2015). *Benin National Action Plan for Energy Efficiency PANEE. (French)*. (<http://documents.worldbank.org/curated/en/2015/07/24819836/eeea-enhancing-world-bank-s-energy-access-investments-through-energy-efficiency>).